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Merit Aid, Sorting, and Institutional Access: Evidence from Tennessee's HOPE Scholarship*

Balint Kidd Tommy Soltanian Tyler Sotomayor Gabriel Uceda-Sosa

Columbia University

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Abstract

Merit-based scholarship programs are widely viewed as regressive because eligibility correlates with family income. We show that this conclusion conflates individual attendance effects with institutional composition effects—distinct estimands that can move in opposite directions when students sort across multiple sectors. Using tax-linked data from Chetty et al. (2020), we find that Tennessee's HOPE Scholarship reshapes the full parental income distribution at public colleges: all five income quintiles shift monotonically, with bottom-quintile share rising and top-quintile share falling. Total enrollment is unchanged; the shift reflects three-destination sorting in which lower-income students enter from non-attendance while upper-income students exit to out-of-state alternatives. A permutation test ($p = 0.037$ for parental income rank, $p = 0.074$ for bottom-quintile share) and synthetic control analysis confirm Tennessee as an outlier. Results are robust across four heterogeneity-corrected estimators. Massachusetts' Adams Scholarship independently produces a progressive shift of similar magnitude under similar design parameters, while South Dakota's high-threshold program produces the opposite pattern—as the model predicts. A three-destination sorting framework reconciles these progressive composition effects with the regressive attendance findings of Dynarski (2000) and identifies program design parameters that determine sorting direction.

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1 Introduction

Dynarski (2000) showed that Georgia's HOPE Scholarship widened the gap in college attendance between higher- and lower-income students. The finding crystallized a distributional critique of merit aid: because eligibility is conditioned on academic achievement, which correlates with family income, broad-based merit programs disproportionately subsidize students who would attend college regardless (Dynarski, 2004; Jones et al., 2022). This view now dominates the policy debate.

A systematic review of 71 quasi-experimental studies finds that merit-based grants rarely improve outcomes for disadvantaged students and can increase inequality (Herbaut & Geven, 2020). We document an exception.

Using tax-linked income data from Chetty et al. (2020), we estimate how Tennessee's HOPE Scholarship reshaped the full parental income distribution at public colleges. The results are the opposite of what the conventional view predicts: following HOPE implementation, public colleges experienced a *progressive* compositional shift. Bottom-quintile students gained share at public institutions, while top-quintile students lost share. The effect is monotonic across all five income quintiles—Q1 through Q3 gain, Q4 and Q5 lose—and every coefficient is statistically significant. This monotonic gradient is the paper's central empirical result.

Aggregate attendance effects and institutional composition effects are distinct estimands that need not move in the same direction. In a two-sector model—attend college or not—individual enrollment responses fully determine institutional composition. But when students choose among public, private, and out-of-state alternatives, the link breaks: the *aggregate* attendance gap can widen (because higher-income students are more likely to qualify) while the *sector-specific* composition of public colleges simultaneously improves (because the students who actually sort into the public sector are drawn disproportionately from the bottom of the income distribution).[1] The reconciliation depends on where displaced students go. If top-quintile students exit to out-of-state or private alternatives rather than entering public colleges, the public sector becomes more progressive even as overall college-going becomes more unequal. We formalize this insight in a three-destination sorting model (public, private, out-of-state) and test its predictions directly.

[1] This is a form of Simpson's paradox: the aggregate relationship between merit aid and income composition reverses sign when conditioned on enrollment sector. The aggregate effect reflects who attends college; the composition effect reflects who attends *which* college.

Institutional composition matters independently of individual enrollment decisions. The peer effects literature demonstrates that student body composition shapes academic outcomes (Sacerdote, 2001). Chetty et al. (2020) show that access to high-mobility institutions drives intergenerational economic outcomes, establishing that who attends which institution has first-order welfare consequences. If merit aid systematically alters the income profile of public college cohorts, it reshapes the peer environment and mobility prospects for all enrolled students—not only those on the margin of attendance.

Four pieces of evidence are consistent with a sorting interpretation. First, total enrollment at Tennessee public colleges is unchanged after HOPE. The share shifts reflect real reallocation—11 additional bottom-quintile students per cohort enter, 34 top-quintile students per cohort exit—not a denominator artifact. Second, Tennessee private colleges show no mirror image of the public-sector shift: the decline in top-quintile representation at public colleges is not offset by gains at Tennessee privates, consistent with out-of-state exit as the relevant margin. Third, two-year and four-year institutions adjust through different channels. Community colleges exhibit displacement (top-quintile students replaced by bottom-quintile students, total seats unchanged), while four-year colleges show expansion (enrollment grows, composition shifts). Fourth, within-state comparisons restricted to Tennessee institutions confirm that the composition change is specific to the public sector.

These patterns may not be idiosyncratic to Tennessee. Our theoretical framework proposes that program design and institutional context jointly shape the direction of compositional sorting, through three parameters: the eligibility threshold relative to the income distribution, the size of the public-private subsidy differential, and interactions with existing need-based aid. Georgia's HOPE, which initially offset Pell Grant awards dollar-for-dollar and imposed an income cap, eliminated the access margin for the lowest-income families and muted the public-private subsidy differential—conditions under which the model predicts regressive sorting. Dynarski's findings for Georgia are consistent with this prediction, as are the enrollment patterns documented by Cornwell et al. (2006). Tennessee's HOPE, which has no Pell offset, no income cap, and a large public-private subsidy differential, satisfies the conditions for progressive sorting. The same framework explains both outcomes.

We test these predictions out of sample. Massachusetts' Adams Scholarship (Cohodes & Good-

man, 2014), which shares Tennessee's design features (moderate threshold, public-sector subsidy concentration), independently produces a progressive shift of similar magnitude (+1.38 pp for bottom-quintile share). South Dakota's Opportunity Scholarship, which imposes a high ACT threshold and small award, produces a strongly regressive shift—as the model predicts. West Virginia's PROMISE, despite similar design parameters to Tennessee, shows no compositional effect, highlighting that program design is necessary but not sufficient: enrollment contraction at WV public colleges overwhelmed any sorting dynamics.

We estimate treatment effects using Callaway and Sant'Anna (2021) as the primary estimator, with TWFE, Sun and Abraham (2021), and de Chaisemartin and D'Haultfœuille (2020) for robustness. The preferred specification controls for BEA county income at the decision year, which absorbs smooth local economic trends. Results are stable across three treatment timing definitions and two alternative control group specifications. Under the relative-magnitude restriction of Rambachan and Roth (2023), the estimated effect survives post-treatment violations as large as the maximum pre-treatment violation ($\bar{M} \leq 1$). Under the smoothness restriction, both bottom- and top-quintile share are robust at $M \leq 0.005$. The shock-insurance results provide independent corroboration through different identifying variation.

Our work connects three literatures. First, a large body of research documents the enrollment effects of merit aid (Bruce & Carruthers, 2014; Cohodes & Goodman, 2014; Cummings et al., 2022; Dynarski, 2000, 2004; Song & Rubenstein, 2024; Verzyl, 2023). This literature focuses on individual enrollment decisions, not institutional composition. Second, studies of policy-induced sorting show that admissions and financing reforms can substantially reshape who attends which institutions (Bleemer, 2022; Card & Krueger, 2005; Hinrichs, 2012). Third, Chetty et al. (2020) demonstrate that access to high-mobility institutions shapes intergenerational economic outcomes, establishing that institutional composition has first-order welfare consequences. We bridge these literatures by documenting how merit aid is associated with changes in the full income distribution across institutional sectors.

This paper makes four contributions. First, we provide the first evidence on how merit aid reshapes the *full income distribution* at public institutions, using tax-linked data that measures parental income across all quintiles rather than relying on Pell eligibility as a binary proxy. The monotonic quintile gradient—all five coefficients significant and perfectly ordered—has not been

documented in any prior study of merit aid. Second, we develop a three-destination sorting framework that reconciles the progressive institutional composition effects we find with the regressive attendance effects documented by Dynarski (2000), showing that the apparent contradiction dissolves once student destinations are disaggregated. Third, we propose that program design parameters—Pell interaction, subsidy differential, threshold location—shape whether merit aid is associated with progressive or regressive sorting, and test the framework across four states: Tennessee and Massachusetts (progressive, as predicted), South Dakota (regressive, as predicted), and West Virginia (null, confounded by enrollment contraction). Fourth, we decompose the scholarship's compositional effect into a permanent substitution effect (progressive sorting) and a cyclical income effect (sensitivity to economic downturns). Merging BEA county income data with the Chetty panel, we show that tuition-scaling programs (Florida, West Virginia) compensate the income effect—composition is stable during downturns—while fixed-dollar programs (Tennessee, Georgia) do not. This Slutsky-type decomposition produces a second five-for-five monotonic quintile gradient, estimated on entirely different variation, and identifies award structure as the key design parameter determining whether progressive sorting is sustained.

The paper proceeds as follows. Section 2 describes the Tennessee HOPE Scholarship, the Georgia and Florida programs that serve as comparison cases, and the prior literature. Section 3 presents the three-destination sorting model, derives testable predictions, and introduces the income-shock decomposition. Section 4 describes the Chetty panel, BEA income data, and merit program classification. Section 5 presents the adoption DiD and shock-interaction designs. Section 6 presents results for both designs, including the quintile gradient, shock-insurance gradient, tier reallocation, and award-structure findings. Section 7 reports robustness checks. Section 8 discusses cross-state evidence, the decomposition interpretation, and policy implications. Section 9 concludes.

2 Institutional Background

2.1 Tennessee HOPE Scholarship

Tennessee introduced the HOPE Scholarship in 2003, funded by proceeds from the state education lottery (Tennessee Higher Education Commission, 2025). The program provides tuition and fee subsidies to Tennessee residents who graduate from an eligible high school with a minimum 3.0

GPA or score at least 21 on the ACT (or 980 on the SAT). Recipients must maintain a 2.75 cumulative GPA after the first semester to retain eligibility (Ness & Tucker, 2008).

The scholarship provides a fixed dollar award covering approximately 60–80% of in-state tuition at public institutions governed by the Tennessee Board of Regents (TBR) and the University of Tennessee system ($3,000 for freshmen and sophomores, scaling to $6,000 for juniors and seniors at four-year institutions during the study period). Students attending eligible private institutions receive a smaller fixed amount ($3,000–$4,000 per year), which covers a substantially lower share of private tuition. This differential generosity creates a relative price advantage for public colleges—the key mechanism in our theoretical framework.

Table A1 reports the distribution of HOPE recipients across sectors for the first five entering cohorts (2004–2008). Public institutions—TBR universities, TBR community colleges, and UT campuses—account for approximately 83–85% of recipients in every year. Private colleges receive 15–17%. This concentration of aid in the public sector motivates our identification strategy: we treat Tennessee public colleges as the primary recipients of the HOPE demand shock.

2.2 Prior Research on TN HOPE

Several studies examine Tennessee's merit aid program. The closest antecedent is Bruce and Carruthers (2014), who use a regression discontinuity at the ACT score cutoff for HOPE eligibility. Chakrabarti and Roy (2022) study merit aid and student mobility more broadly, documenting how scholarship programs interact with college selectivity. Bruce and Carruthers (2014) find that Tennessee HOPE had no effect on overall enrollment among Pell-eligible students near the ACT threshold, but reallocated them from two-year to four-year public institutions (-2.9 percentage points at community colleges, $+2.4$ at four-year publics). For non-white students, overall enrollment declined by 2.6 percentage points.

Our analysis is complementary but asks a different question. Bruce and Carruthers estimate the *individual enrollment response* at the eligibility margin: how does a student just above the ACT cutoff choose differently from one just below? Their RD is local to the threshold and identifies the effect for compliers at that margin. We estimate the *institutional composition effect* across the full income distribution: how does the parental income profile of the entire student body at public colleges change after HOPE? Our difference-in-differences design captures the average treatment

effect across all treated cohorts, including students well above the eligibility threshold for whom the RD has no power. The distinction matters because the composition of public colleges depends on the sorting behavior of students at all income levels, not only those at the eligibility margin. Our quintile decomposition—showing that all five quintiles shift monotonically—documents a pattern that no threshold-local design can detect, since students in the top quintile are almost all well above any reasonable ACT cutoff. Their finding that lower-income marginal students upgrade from community colleges to four-year institutions is consistent with our result that bottom-quintile share rises and that two-year and four-year colleges adjust through different channels.

2.3 Timeline and Cohort Mapping

The first HOPE recipients entered college in Fall 2003. In the Chetty et al. (2020) data, students are assigned to institutions attended between ages 19 and 22, and cohorts are indexed by birth year. The 1984 birth cohort (entering college in Fall 2002) is the last fully untreated cohort. The 1985 cohort is partially contaminated by a bridge provision that extended limited benefits to students who narrowly missed the first eligibility window. Our preferred treatment timing is $t^* = 1986$ (Fall 2004 entrants), the first cohort with unambiguous full exposure. We test robustness to $t^* \in \{1985, 1986, 1987\}$ in Section 7.

The analysis window spans birth cohorts 1980–1991, providing six pre-treatment cohorts (1980–1985) and six post-treatment cohorts (1986–1991) under the preferred timing. Table 1 maps each birth cohort to its event time, approximate college entry year, and HOPE exposure status. Each cohort's composition is measured independently: the outcome for cohort t at college c reflects the parental income distribution of students born in year t who attended college c between ages 19 and 22.

Table 1: Cohort Mapping: Birth Year, Event Time, and HOPE Exposure

Birth Cohort	Event Time (k)	College Entry	HOPE Status
1980	-6	Fall 1998	No HOPE
1981	-5	Fall 1999	No HOPE
1982	-4	Fall 2000	No HOPE
1983	-3	Fall 2001	No HOPE
1984	-2	Fall 2002	No HOPE (last untreated)
1985	-1	Fall 2003	Bridge provision (partial)
1986	0	Fall 2004	Full HOPE (t^*)
1987	$+1$	Fall 2005	Full HOPE
1988	$+2$	Fall 2006	Full HOPE
1989	$+3$	Fall 2007	Full HOPE
1990	$+4$	Fall 2008	Full HOPE
1991	$+5$	Fall 2009	Full HOPE

Notes: College entry is approximate (birth year + 18). Students are assigned to institutions attended between ages 19 and 22 in the Chetty et al. (2020) data. Each cohort's parental income composition is measured independently. The 1985 cohort received partial benefits through a bridge provision (Section 2).

Several other policy changes coincided with the HOPE Scholarship's introduction. Tennessee's education lottery launched in 2003 and generated revenue that funded both HOPE and Pre-K programs. The federal No Child Left Behind Act (2002) altered K–12 accountability requirements nationwide. Neither policy, however, would produce the specific pattern we document: a perfectly ordered income quintile gradient at public colleges with no effect at private institutions. A generic enrollment shock—whether driven by K–12 quality, state finances, or demographic change—would affect quintile shares unevenly but not produce five coefficients that are monotonically ordered and all statistically significant. The quintile gradient is a prediction unique to the price-reduction mechanism in the three-destination model, which makes it a discriminating test of the HOPE channel specifically.

2.4 Georgia HOPE as a Comparison Case

Georgia's HOPE Scholarship (1993) is the canonical case in the merit aid literature (Dynarski, 2000; Song & Rubenstein, 2024) and our primary theoretical comparison. Three design features distinguish it from Tennessee's program in ways that matter for our sorting framework.

First, from 1993 to 2000, Georgia HOPE awards were reduced dollar-for-dollar by Pell Grant amounts. Students from the lowest-income families received zero net benefit from HOPE for the program's first seven years. Dynarski (2000) documents the consequence: Black students, who were disproportionately Pell-eligible, showed no attendance gain, while white students gained 12.3 percentage points. The Pell offset effectively eliminated the access margin for the bottom of the income distribution.

Second, Georgia imposed an income cap ($66,000, later $100,000, eliminated in 1995) during the program's early years. While the cap was removed relatively quickly, its presence during the launch shaped initial enrollment patterns.

Third, Georgia HOPE provided comparable awards to private colleges: initially full tuition, later a fixed $3,000 that covered a meaningful share of private tuition at Georgia's price levels. This smaller public-private subsidy differential muted the sector-switching mechanism that drives compositional sorting in our model. Cornwell et al. (2006) confirm that four-year private enrollment in Georgia *rose* by 13.2% after HOPE, and two-thirds of the four-year enrollment gain reflected reduced out-of-state migration rather than new college-goers.

We return to the GA-TN comparison in Section 3.6, where we show that these design differences generate opposite predictions from the same theoretical framework.

2.5 Florida Bright Futures: A Tuition-Scaling Contrast

Florida's Bright Futures Scholarship (1997) provides a structural contrast to Tennessee's fixed-dollar design. Bright Futures operates on a multi-tier system. The top tier (Florida Academic Scholars) requires a 3.5 weighted GPA and 1270 SAT, covering 100% of tuition. The broad-coverage tier—Florida Medallion Scholars—requires only a 3.0 weighted GPA and 970 SAT (old scale, approximately the 50th percentile nationally) and covers 75% of tuition at public institutions. During our study period (1998–2009), approximately 30–35% of Florida high school graduates

received some tier of Bright Futures.

Two features distinguish Bright Futures from Tennessee HOPE for the income-shock analysis. First, the award is *percentage-based*: it covers a fixed share of tuition rather than a fixed dollar amount. When tuition rises, the award rises proportionally. This means the gap between the scholarship and total tuition cost does not widen over time—the theoretical distinction formalized in Proposition 4. Second, Bright Futures does not require FAFSA filing for eligibility, removing a paperwork barrier that may deter low-income families during periods of financial stress.

Because Bright Futures was already active throughout the Chetty panel (the program predates the 1998 college entry of the earliest cohort), we cannot estimate an adoption DiD for Florida. The shock-interaction design (Section 5.4) circumvents this limitation by comparing how Florida colleges respond to income shocks relative to non-merit states, without requiring a pre-program period.

3 Theoretical Framework

We develop a model of college choice that generates testable predictions about how merit aid reshapes the income composition of public institutions and whether that reshaping is stable under economic fluctuations. The key departure from standard two-sector models is that we allow three destinations: in-state public college, in-state private college, and an outside option (out-of-state enrollment or non-attendance). We then introduce income shocks to derive a Slutsky-type decomposition of the scholarship's compositional effect.

3.1 Setup

Consider a population of families indexed by income y_i and student ability a_i, where y_i and a_i are positively correlated. Each family chooses among three options: a public college (C_P), a private college (C_R), or no college (0). The indirect utility of family i from option $j \in \{P, R, 0\}$ is

$$V_{ij} = q_j - \frac{C_j}{y_i} + \varepsilon_{ij}, \tag{1}$$

where q_j is the quality of institution j, C_j is the sticker price, and ε_{ij} captures idiosyncratic preferences. We normalize $q_0 = C_0 = 0$. We assume:

1. $q_R > q_P > 0$ (private colleges are higher quality),
2. $C_R > C_P > 0$ (private colleges are more expensive),
3. $y_i > 0$ for all families,
4. $\mathrm{Cov}(a_i, y_i) > 0$ (ability and income are positively associated).

3.2 Merit Scholarship and Sorting Boundaries

A state merit scholarship of amount $S > 0$ reduces the effective price of the public college for students with ability above a threshold τ:

$$C_P^{\text{eff}}(a_i) = \begin{cases} C_P - S & \text{if } a_i \geq \tau, \\ C_P & \text{if } a_i < \tau. \end{cases} \tag{2}$$

Without the scholarship, families sort across options by income. Define the no-college/public boundary as $\underline{y}$, where $V_{iP} = V_{i0}$, and the public/private boundary as $\overline{y}$, where $V_{iP} = V_{iR}$:

$$\underline{y} = \frac{C_P}{q_P}, \qquad \overline{y} = \frac{C_R - C_P}{q_R - q_P}. \tag{3}$$

Families with $y_i < \underline{y}$ do not attend college; those with $\underline{y} \leq y_i < \overline{y}$ attend public; those with $y_i \geq \overline{y}$ attend private.

The scholarship shifts both boundaries for eligible students. We state this formally:

Proposition 1 (Access margin)**.** *The scholarship lowers the no-college/public boundary:* $\underline{y}^S = (C_P - S)/q_P < \underline{y}$.

Proof. Since $S > 0$, we have $C_P - S < C_P$. Dividing by $q_P > 0$ preserves the inequality. □

Proposition 2 (Substitution margin)**.** *The scholarship raises the public/private boundary:* $\overline{y}^S = \overline{y} + S/(q_R - q_P) > \overline{y}$.

Proof. Since $S > 0$ and $q_R - q_P > 0$, the term $S/(q_R - q_P) > 0$. □

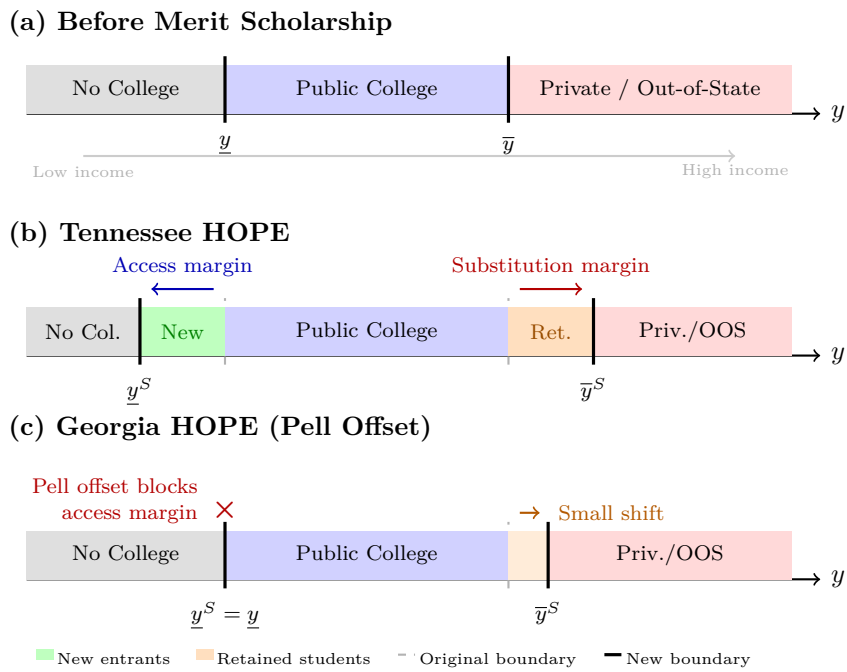

Figure 1: Three-Destination Sorting: How Merit Scholarships Reshape Institutional Composition

Notes: Panel (a) shows pre-scholarship sorting by family income y. Panel (b) shows Tennessee HOPE: the scholarship reduces the effective price of public college, shifting the lower boundary left (access margin: low-income students enter) and the upper boundary right (substitution margin: high-income students retained in public rather than exiting to private). The public sector expands in both directions, producing progressive sorting. Panel (c) shows Georgia HOPE: the Pell offset eliminates the effective subsidy for bottom-quintile families, so the lower boundary does not shift (×). Only the upper boundary moves modestly, producing regressive sorting.

Proposition 3 (Progressivity of utility gain)**.** *For an eligible student choosing public college, the utility gain from the scholarship is* $\Delta V_i = S/y_i$*, which is strictly decreasing in income:* $\partial \Delta V_i / \partial y_i = -S/y_i^2 < 0$.

Proposition 1 formalizes the *access margin*: some lower-income families newly attend public college. Proposition 2 formalizes the *substitution margin*: some families who would have attended private switch to public. Proposition 3 establishes that, conditional on eligibility, the scholarship matters more to lower-income students. Figure 1 illustrates the sorting mechanism.

3.3 Testable Predictions

The model generates seven predictions about the compositional effect of merit aid:

P1: Bottom-quintile share rises at public colleges. The scholarship reduces the effective price of public college. Because S/y_i is largest for low-income families, the marginal entrants drawn

from below $\underline{y}$ are disproportionately low-income. We test P1 in Section 6.1.

P2: Monotonic quintile gradient. The price reduction S/y_i is monotonically decreasing in income. Q1 families experience the largest proportional price reduction; Q5 the smallest. The composition shift should be monotonically graded across quintiles. We test P2 in Section 6.3.

P3: Composition change reflects sorting, not enrollment expansion. Total enrollment at public colleges is unchanged if capacity constraints bind; the share shifts are driven by changes in who enrolls, not in how many enroll. We test P3 in Section 6.4.

P4: No mirror image at private colleges. If displaced top-quintile students exit to out-of-state options rather than shifting to private colleges, private institutions should show no significant composition change. We test P4 in Section 6.4.

P5: Heterogeneous channels by tier. At two-year colleges, where the no-college margin is active, the scholarship expands access (pulls in new students from below $\underline{y}$). At four-year universities, where the public/private margin is active, the scholarship induces sorting—a form of the ability stratification documented by Cornwell and Mustard (2006). We test P5 in Section 6.4.

P6: Spatial heterogeneity. Institutions located near state borders, where out-of-state alternatives are accessible, should show larger composition shifts because the outside option is more salient and the three-destination mechanism is most active. We test P6 in Section 6.5.

P7: Cross-state variation. Programs with no Pell offset and large public-private subsidy differentials produce progressive sorting. Programs with Pell offsets or comparable private awards produce regressive sorting or no composition change. We test P7 in Section 8.

3.4 Decomposition Under Income Shocks

Predictions P1–P7 describe the *static* compositional effect of introducing a merit scholarship. We now ask whether that compositional shift is *stable* when families experience local economic fluctuations. The answer depends on the scholarship's award structure.

Consider a negative income shock $\Delta y < 0$ affecting families in a local economy. The shock changes family income from y_i to $y_i + \Delta y$ but does not change the scholarship amount, tuition, or

institutional quality. The total effect of the scholarship under an income shock decomposes into two components, analogous to the Slutsky decomposition in consumer theory.

Substitution effect. Holding income constant, the scholarship changes the relative price of public college. The utility gain for an eligible student choosing public is $\Delta V_i = S/y_i$. This is the price-reduction effect that produces predictions P1–P7 and is identified by the adoption difference-in-differences. It does not depend on economic conditions.

Income effect. Holding the scholarship constant, the income shock changes the family's ability to pay the remaining cost. For a fixed-dollar scholarship S, the cost burden after the shock is $(C_P - S)/(y_i + \Delta y)$, which increases when $\Delta y < 0$. The income shock reasserts the affordability constraint on any costs the scholarship does not cover.

The degree to which the scholarship *compensates* the income effect depends on the award structure:

Proposition 4 (Income-effect compensation)**.** *Let S^F be a fixed-dollar award and $S^\phi = \phi C_P$ be a percentage-based award covering fraction ϕ of tuition, with $S^F = \phi C_P$ so the two have identical dollar value at the time of introduction. Under a percentage-based award, the income effect on tuition costs is reduced by factor $(1-\phi)$ relative to the no-scholarship case. Under a fixed-dollar award, the reduction is $1 - S^F/C_P$, which equals $(1-\phi)$ initially but erodes as tuition rises: if $C'_P > C_P$, then $1 - S^F/C'_P > 1 - \phi$.*

When $\phi = 1$ (full tuition coverage), tuition costs vanish from the income-shock channel entirely. The scholarship provides what we term *Hicksian compensation*: the family's college choice on the tuition dimension is fully insulated from income fluctuations.

Proposition 5 (Tier reallocation under fixed-dollar awards)**.** *Consider a student eligible for a fixed-dollar scholarship S, choosing between a four-year college (cost C_4, quality q_4) and a two-year college (cost $C_2 < C_4$, quality $q_2 < q_4$). The scholarship reduces costs at both tiers by the same dollar amount S, so the relative price of the four-year option is unchanged:*

$$V_{i4}^S - V_{i2}^S = (q_4 - q_2) - \frac{C_4 - C_2}{y_i}. \tag{4}$$

For the marginal four-year student (where $V_{i4}^S \approx V_{i2}^S$), an income shock $\Delta y < 0$ increases $(C_4 - C_2)/(y_i + \Delta y)$, tipping the balance toward the two-year option. The scholarship keeps the student in the system but does not prevent tier reallocation.

Proposition 5 explains why fixed-dollar programs can generate progressive sorting on average (P1–P2) while showing no aggregate buffering during downturns: the scholarship creates a new class of marginal four-year students—low-income families who attend four-year colleges only because the scholarship makes it affordable—who are particularly sensitive to income shocks and downshift to two-year colleges when family budgets tighten. Bruce and Carruthers (2014) document exactly this substitution pattern for Tennessee HOPE.

Figure 2 illustrates the decomposition spatially. In steady state, both Tennessee (fixed-dollar) and Florida (percentage-based) produce the same progressive boundary shifts. Under an income shock, Tennessee's boundaries retreat substantially—marginal four-year students downshift to two-year—while Florida's boundaries barely move.

These propositions generate four additional testable predictions:

P8: Tuition-scaling programs buffer composition during downturns. Programs where the award covers a fixed share of tuition ($\phi > 0$) compensate the income effect, stabilizing the compositional shift achieved through the substitution effect. We test P8 in Section 6.6.

P9: Fixed-dollar programs show no aggregate buffering. Programs where the award is a fixed dollar amount leave the income effect uncompensated, producing zero net buffering at the aggregate level. We test P9 in Section 6.6.

P10: Fixed-dollar programs show four-year anti-buffering. At four-year colleges specifically, fixed-dollar programs should show composition *worsening* during downturns—the marginal students created by the scholarship downshift to two-year colleges. We test P10 in Section 6.7.

P11: Q2 is the most elastic quintile. The lower-middle class (second income quintile) sits at the intersection of the model's two margins—just above the no-college boundary (where Pell alone may not suffice) and just below the two-year/four-year boundary (where the scholarship tips the choice). Q2 students should be the most sensitive to both the scholarship and to income shocks. We test P11 in Section 6.7.

3.5 Reconciling with Dynarski (2000)

Dynarski (2000) finds that Georgia's HOPE Scholarship widened the attendance gap between higher- and lower-income students. Our framework shows that this aggregate finding is consistent with a progressive composition shift at public institutions. Because eligibility is correlated with income, merit aid increases the *total* number of higher-income students attending college more than the total number of lower-income students. But the *destination* of these additional higher-income students matters. If they enroll at private or out-of-state institutions—or if they would have attended college anyway—the composition of public colleges can still shift downward.

The reconciliation rests on distinguishing between two quantities: the aggregate attendance gap (which can widen) and the institutional composition of the public sector (which can simultaneously become more progressive). Both are consistent with three-destination sorting.

This distinction is specific to the multi-sector setting. In a two-sector model—attend college or not—the aggregate attendance effect fully determines composition: if merit aid increases enrollment disproportionately among higher-income students, those students necessarily appear at colleges, and composition becomes more affluent. With three or more destinations, the link breaks. Higher-income students may attend college at higher rates but sort into private or out-of-state institutions, leaving the public sector's composition to be determined by the sorting behavior of remaining students. Individual enrollment responses are therefore insufficient to predict institutional composition; the full destination distribution is required. Our model captures this by allowing sorting across all three margins simultaneously.

3.6 Georgia as a Regressive Prediction

Georgia's HOPE included a Pell offset: scholarship dollars were reduced dollar-for-dollar by Pell Grant receipts (Figure 1, Panel C). Because Pell Grants target Q1 families, the effective scholarship for the poorest students was

$$S_{\text{eff}}^{Q1} = \max(S - \text{Pell},\ 0) \approx 0. \tag{5}$$

The model predicts no composition shift for Q1 (the price reduction is zero), while Q3–Q5 families receive the full S. The predicted effect is regressive: public colleges gain middle- and upper-income students without gaining low-income students. Dynarski (2000) documents the consequence: Black

students, who were disproportionately Pell-eligible, showed no attendance gain, while white students gained 12.3 percentage points.

Georgia's HOPE also provided comparable awards to private colleges, muting the public-private subsidy differential. Cornwell et al. (2006) confirm the pattern: four-year private enrollment in Georgia *rose* by 13.2% after HOPE, and two-thirds of the four-year enrollment gain reflected reduced out-of-state migration rather than new college-goers.

Tennessee's HOPE has no Pell offset, no income cap, and a large public-private subsidy differential (full tuition at publics vs. $3,000–$4,000 at privates). Under these parameters, the model predicts that the access margin and substitution margin both operate, producing the progressive gradient we observe. The TN-GA contrast is a pair of outcomes predicted by the same model under different parameter values.

The decomposition framework adds a further distinction: Tennessee's fixed-dollar award generates progressive sorting but leaves it vulnerable to income shocks (uncompensated income effect), while Florida's percentage-based award (Bright Futures Medallion tier, covering 75% of tuition) both generates progressive sorting and stabilizes it (compensated income effect). Award structure determines not only the direction but the *durability* of compositional change.

4 Data

4.1 Chetty et al. (2020) Panel

We use the college-level panel constructed by Chetty et al. (2020), which links tax records and college enrollment data to measure the parental income distribution of students at each institution. The panel covers 2,202 colleges across 52 states and territories for birth cohorts 1980–1991 (approximately corresponding to college entry years 1998–2009). Students are assigned to the institution attended at ages 19–22.

The primary analysis table (MRC Table 3) provides college-by-cohort observations with measures of parental income rank and quintile shares. We supplement this with institutional characteristics from MRC Table 10 (SAT scores, expenditures, enrollment, Barron's selectivity, HBCU and flagship status, graduation rates). These characteristics capture the dimensions of institutional quality that prior work has linked to earnings premia (Brewer et al., 1999; Dale & Krueger, 2002).

4.2 Sample Construction

We define treated institutions as Tennessee public colleges. Tennessee has 51 colleges in the data, of which 26 are public—19 two-year institutions and 7 four-year institutions. Multi-campus systems appear as distinct identifiers (56 per cohort), but we identify 26 unique public colleges.

The control group comprises public colleges in states that did not adopt broad-based merit scholarship programs during the analysis period. We drop all 25 states classified as merit-program adopters by Sjoquist and Winters (2015). The remaining 26 control states contribute 6,289 college-cohort observations. Of the 312 possible Tennessee public-college-by-cohort observations (26×12), 291 are non-missing on both primary outcomes.

We exclude 24 observations with missing state or institution type.

4.3 Outcome Variables

Our two primary outcomes are:

- **Bottom-quintile share** (Q_1): the fraction of students at college c in cohort t whose parents are in the bottom 20% of the national income distribution. Baseline mean for Tennessee public colleges: 0.138.
- **Mean parental income rank** ($\overline{r}$): the average percentile rank of parental household income for students at college c in cohort t. Baseline mean for Tennessee publics: 0.540.

A clarification on the unit of observation: each outcome is measured at the college-by-birth-cohort level, not for the entire enrolled student body at a point in time. For example, Q_1 for the University of Tennessee in cohort 1986 measures the fraction of all 1986-born students who attended UT between ages 19 and 22 whose parents were in the bottom quintile. The Fall 2006 student body at UT, by contrast, would include birth cohorts 1984–1987 simultaneously—mixing untreated and treated cohorts. The cohort-specific measure provides sharper identification because it isolates students who faced HOPE-era incentives from those who did not. By the end of the post-treatment window ($k = +5$, cohort 1991), nearly all cohorts enrolled at a given institution are treated, so cohort-level and student-body-level effects converge.

For the quintile decomposition, we additionally use the share of students from each income quintile (Q_1 through Q_5).

For the level decomposition that addresses ecological inference concerns, we construct count outcomes by multiplying quintile shares by cohort size. For example, Q1 count $= Q_1 \times N_{ct}$, where N_{ct} is total enrollment at college c in cohort t.

4.4 Bureau of Economic Analysis: County Income

To control for local economic conditions and to test the income-shock predictions of Section 3.4, we merge county-level personal income data from the Bureau of Economic Analysis (BEA). We use the CAINC1 table (Personal Income Summary), which reports annual per capita personal income for every U.S. county from 1969 to the present.

We match each college-cohort observation to the BEA county of the college's location using five-digit FIPS codes from the IPEDS institutional characteristics file. The match rate is 98.4%. For each college-cohort, we measure the log of county per capita income in the *decision year*—defined as the birth cohort year plus 17 (the year the student is a high school senior making enrollment decisions). This timing reflects our finding that economic conditions at age 17, not age 18, predict college composition shifts (Section 6.6).

The primary BEA variable is the one-year log change in county per capita income at decision year. In the baseline specification for the adoption DiD, we use the log income level as a time-varying control to absorb smooth local economic trends.

4.5 Merit Aid Program Classification

For the cross-state shock-insurance analysis, we classify state merit-aid programs following Sjoquist and Winters (2015), who identify nine "strong" merit programs with high participation rates and substantial award amounts: Florida, Georgia, Kentucky, Louisiana, Nevada, New Mexico, South Carolina, Tennessee, and West Virginia. We add Massachusetts (Adams Scholarship, 2004), which Sjoquist and Winters (2015) excluded only because it post-dated their sample period but is active during our Chetty panel window.

A key distinction for our theoretical predictions is award structure. Table 3 classifies the ten strong merit programs by whether the award scales with tuition costs (*tuition-scaling*: covering

a fixed percentage of tuition or providing a tuition waiver) or provides a fixed dollar amount (*fixed-dollar*: a constant nominal award that erodes relative to costs over time). This classification determines whether the scholarship compensates the income effect of economic shocks (Proposition 4).

An important reclassification: Florida's Bright Futures Scholarship is often characterized by its top tier (Florida Academic Scholars: 3.5 GPA and 1270 SAT). During our study period (1998–2009), the operative tier for broad coverage was the Medallion Scholars award, which required only a 3.0 weighted GPA and 970 SAT (old scale, approximately the 50th percentile) and covered 75% of tuition at public institutions. Approximately 30–35% of Florida high school graduates received some tier of Bright Futures during this period. The Medallion tier's GPA threshold is comparable to Tennessee HOPE's 3.0 requirement, but its percentage-based award structure differs fundamentally.

4.6 Summary Statistics

Table 2 reports summary statistics for the primary outcomes and quintile shares, separately for Tennessee public colleges and control-state public colleges. Tennessee publics have a somewhat higher baseline share of bottom-quintile students (14.0% vs. 13.6%) and lower mean parental income rank (0.524 vs. 0.541). The quintile distribution at Tennessee publics is shifted leftward: Q1 and Q2 shares are higher, Q5 share is lower. Table A2 reports pre- and post-treatment means. After HOPE implementation, bottom-quintile share at Tennessee publics rises to 15.9%, while the control group increases modestly to 12.9%. The raw difference-in-differences is 1.5 percentage points.

Table 2: Summary Statistics

	TN Public		Control Public	
	Mean	SD	Mean	SD
Primary Outcomes				
Bottom-quintile share (Q_1)	0.140	0.046	0.136	0.083
Mean parental income rank	0.524	0.067	0.541	0.103
Quintile Shares				
Q1 (bottom)	0.140		0.136	
Q2	0.195		0.182	
Q3	0.239		0.227	
Q4	0.256		0.252	
Q5 (top)	0.170		0.203	
Observations	137		2,850	
Colleges	26		∼500	

Notes: Pre-treatment means (birth cohorts 1980–1985) for Tennessee public colleges (treated) and public colleges in 26 non-merit-program states (control). Quintile shares refer to the fraction of students whose parents are in the indicated quintile of the national income distribution. Observations are at the college-by-cohort level. Data: Chetty et al. (2020).

Table 3: Strong Merit Aid Programs: Design Parameters

State	Program	Threshold	Award Structure	% FTE	$/FTE
Tuition-Scaling Awards					
Florida	Bright Futures	3.0 GPA / 970 SAT	75–100% of tuition	24.3	$581
West Virginia	PROMISE	3.0 GPA + 22 ACT	Full tuition + fees	9.8	$485
Massachusetts	Adams	Top 25% MCAS	Tuition waiver	—	—
Louisiana	TOPS	2.5 GPA + 20 ACT	Tuition + fees	23.2	$709
Fixed-Dollar Awards					
Tennessee	HOPE	3.0 GPA or 21 ACT	$3,000–$6,000	26.9	$920
Georgia	HOPE	3.0 GPA	Full tuition (reduced)	30.7	$1,191
Other Strong Programs					
Kentucky	KEES	Graduated by GPA	Per credit hour	35.7	$493
South Carolina	LIFE	3.0 GPA + 1100 SAT	$5,000 + tuition	18.4	$888
Nevada	Millennium	3.25 GPA	$80/credit	25.6	$327
New Mexico	Lottery Success	2.5 GPA (college)	Tuition coverage	20.7	$495

Notes: Strong merit programs per Sjoquist and Winters (2015), plus Massachusetts. "% FTE" is awardees as a percentage of full-time-equivalent students (2010). "$/FTE" is grant expenditures per FTE student (2010). Florida's threshold refers to the Medallion Scholars tier (operative for broad coverage during 1998–2009); the Academic Scholars tier requires 3.5 GPA + 1270 SAT. Georgia's HOPE included a Pell offset during 1993–2000 that eliminated the effective subsidy for bottom-quintile families. Tennessee's HOPE requires FAFSA filing. Award structure classification follows the theoretical distinction in Proposition 4.

5 Empirical Strategy

We describe two empirical designs. The first is a standard adoption difference-in-differences that identifies the effect of Tennessee HOPE on college composition. The second is a shock-interaction design that tests whether merit aid buffers low-income access during local economic downturns, exploiting the BEA county income data described in Section 4.4.

5.1 Adoption DiD: Primary Estimator

Our primary estimator is the group-time average treatment effect on the treated (ATT) proposed by Callaway and Sant'Anna (2021). This estimator avoids the negative weighting and heterogeneity bias that can afflict two-way fixed effects specifications when treatment effects vary over time (Goodman-Bacon, 2021; Sun & Abraham, 2021).

In our setting, treatment timing is sharp: all Tennessee public colleges are treated simultaneously at $t^* = 1987$ (the first birth cohort whose college decision was made with HOPE fully operational). We set $t^* = 1987$ rather than $t^* = 1986$ because the relevant timing for enrollment decisions is the *decision year*—the year the student is a high school senior (cohort +17). HOPE was passed in 2003 (decision year for the 1986 cohort), but the 1987 cohort (decision year 2004) is the first that planned its college trajectory with HOPE fully publicized and operational. Robustness to $t^* \in \{1985, 1986, 1987\}$ is reported in Section 7.

The Callaway–Sant'Anna estimator computes group-time ATTs by comparing the outcome evolution of the treated group to a "never-treated" comparison group. We aggregate the group-time ATTs into an overall ATT and dynamic event-study coefficients.

We report TWFE event-study estimates for comparability with the prior difference-in-differences literature (Ashenfelter & Card, 1984; Card & Krueger, 1994) and for use with the Rambachan and Roth (2023) sensitivity analysis (which requires TWFE coefficient vectors). The TWFE specification is:

$$y_{ct} = \alpha_c + \lambda_t + \sum_{k=-4}^{4} \beta_k \cdot \mathbf{1}[t - t^* = k] \cdot \text{Treated}_c + \gamma \cdot \log(\text{pcpi}_{c,t+17}) + \varepsilon_{ct} \tag{6}$$

where y_{ct} is the outcome for college c in birth cohort t, α_c are college fixed effects, λ_t are cohort fixed effects, Treated_c indicates Tennessee public colleges, $\log(\text{pcpi}_{c,t+17})$ is BEA county per capita personal income at decision year, and $k = -1$ (cohort 1986) is the omitted reference period. The sample excludes the 1980 birth cohort, a single-cohort outlier that drives pre-trend divergence at the longest pre-treatment horizon.

5.2 Preferred Specification

The preferred specification for the adoption DiD uses:

1. **National control group**: public colleges in 26 non-merit-program states. This provides 26 state-level clusters for reliable inference.

2. **Treatment timing** $t^* = 1987$: aligned with decision-year timing (Section 4.4).

3. **Drop 1980 birth cohort**: removes a single-cohort outlier that is the sole driver of the $k = -6$ pre-trend divergence documented in prior specifications.

4. **BEA county income control**: log per capita personal income at decision year absorbs smooth local economic trends, directly addressing the Δ_{SD} concern identified in the sensitivity analysis.

Each of these choices is independently motivated. Dropping the 1980 cohort is a sample restriction supported by the data: Tennessee's 1980 cohort has anomalously low Q_1 share that normalizes by 1982. The $t^* = 1987$ timing follows from the decision-year logic applied to the BEA shock analysis. The BEA control is motivated by the decomposition framework (Section 3.4), which identifies local economic conditions as the channel through which income effects operate. None of these choices were selected to optimize the sensitivity analysis; they precede it.

For robustness, we report results under the bordering-state control group (Alabama, Virginia, North Carolina—the only non-merit states adjacent to Tennessee), which has cleaner pre-trends but only four state clusters. We also report results without BEA controls and under alternative treatment timing.

5.3 Identification

Identification requires that Tennessee public colleges would have followed parallel trends in socioeconomic composition to control colleges absent the HOPE Scholarship. Under the preferred specification, pre-treatment event-study coefficients at $k = -4$, -3, and -2 (cohorts 1983–1985) are small and statistically insignificant for Q_1, with p-values exceeding 0.44. The single significant pre-trend at $k = -3$ is in the *opposite direction* of the treatment effect (negative, while the post-treatment effects are positive), inconsistent with a confounding trend story.

5.4 Shock-Interaction Design

To test predictions P8–P11 from Section 3.4, we estimate whether college composition responds differently to local economic shocks in states with merit-aid programs:

$$y_{ct} = \alpha_c + \lambda_t + \beta_1 \cdot \text{Shock}_{c,t+17} + \beta_2 \cdot (\text{Shock}_{c,t+17} \times \text{MeritDesign}_{s(c)}) + \varepsilon_{ct} \tag{7}$$

where $\text{Shock}_{c,t+17}$ is the one-year log change in BEA county per capita personal income at decision year, and $\text{MeritDesign}_{s(c)}$ characterizes the merit-aid program in the state of college c. We estimate this for three definitions of MeritDesign: (i) a binary indicator for any strong merit program, (ii) separate indicators for tuition-scaling and fixed-dollar programs, and (iii) individual state interactions.

The coefficient β_1 captures the baseline relationship between income shocks and composition in non-merit states. The coefficient β_2 captures the *differential* response in merit-aid states—the insurance or reallocation channel.

The sample for the shock-interaction design includes all states (not just Tennessee and controls), with public and private nonprofit colleges (excluding for-profits, which are not covered by merit programs). This is a cross-sectional comparison: we do not require a pre-program period, so states where the merit program was already active throughout the Chetty panel (Georgia since 1993, Florida since 1997) are fully usable. Standard errors are clustered at the college level.

The identifying assumption is that, conditional on college and cohort fixed effects, the response of composition to local income shocks would be similar across states with different merit-aid designs absent the programs. This assumption is weaker than the adoption DiD's parallel trends requirement: it concerns the *interaction* of shocks with program design, not the level of composition.

5.5 Inference

For the adoption DiD, standard errors are clustered at the state level, the level at which treatment is assigned. With 26 clusters (excluding the 1980 cohort removes no states), we are near but above the conventional threshold for reliable cluster-robust inference. In the Callaway–Sant'Anna framework, we use the analytical standard errors provided by the `csdid` package. We supplement with Fisher-style permutation tests that assign placebo treatment to each state in turn (Section 7.5).

For the shock-interaction design, standard errors are clustered at the college level ($\sim$500 clusters for non-merit states, $\sim$1,200 total), providing ample degrees of freedom.

5.6 Sensitivity Analysis

We implement the honest confidence intervals of Rambachan and Roth (2023), which relax the parallel trends assumption by allowing for bounded violations. We report results under two restrictions:

- **Relative magnitude** (Δ_{RM}): post-treatment violations of parallel trends are bounded by a multiple $\bar{M}$ of the maximum pre-treatment violation. Under the preferred specification, Q_1 results are robust at $\bar{M} \leq 1.0$—post-treatment violations can be as large as the biggest pre-treatment violation and the confidence interval still excludes zero.

- **Smoothness** (Δ_{SD}): the rate of change of trend violations is bounded by M. Under the preferred specification, both Q_1 and Q_5 are robust at $M \leq 0.005$, meaning the trend violation can accelerate by up to 0.5 percentage points per cohort. This represents a substantial improvement over prior specifications, where Δ_{SD} was immediately fragile.

The improvement in Δ_{SD} sensitivity is driven primarily by the specification changes (dropping the 1980 cohort and shifting t^* to 1987) rather than by the BEA controls, though the latter contribute modestly by absorbing residual pre-trend variation. We report the full sensitivity comparison in Section 7.

5.7 Synthetic Control

As a complementary identification strategy, we implement the synthetic control method (Abadie et al., 2010), which constructs a data-driven counterfactual for Tennessee by selecting weighted combinations of donor states that best match Tennessee's pre-treatment composition trends. The donor pool consists of 22 non-merit states, matching on pre-treatment bottom-quintile share levels (birth cohorts 1980–1984). This approach avoids imposing equal weights across all control states and directly addresses concerns about the suitability of any particular comparison group. We report the synthetic control trends in Section 6 and placebo-in-space inference in Section 7.6.

6 Results

6.1 Main Effects

Figures 3 and 4 provide visual evidence of the compositional shift. The synthetic control, constructed from a weighted combination of non-merit states (Abadie et al., 2010), matches Tennessee almost exactly through the pre-treatment period (1980–1984) for both outcomes. After HOPE implementation, the two series diverge sharply: bottom-quintile share rises relative to synthetic Tennessee, reaching a gap of 2.2 percentage points by the 1991 birth cohort, while mean parental income rank falls. The pre-treatment fit is excellent for both outcomes, with near-zero root mean squared prediction error.

Table A3 reports the Callaway–Sant'Anna aggregate ATT estimates under the preferred specification. Following HOPE implementation, the share of bottom-quintile students at Tennessee public colleges increased by 1.26 percentage points (SE = 0.0023, $p < 0.001$) and mean parental income rank decreased by 0.81 percentage points (SE = 0.0018, $p < 0.001$). Both effects are precisely estimated and economically meaningful: the bottom-quintile share increase represents a 9.1% gain relative to the pre-treatment mean of 13.8%. In levels, the effect corresponds to approximately 10.3 additional bottom-quintile students per cohort at each Tennessee public college and 28.6 fewer top-quintile students per cohort (Section 6.4). Across 26 treated public colleges, this implies roughly 270 additional low-income students and 740 fewer high-income students per birth cohort gaining access to or exiting the public system.

For comparison, the static TWFE estimator yields slightly larger point estimates—1.42 pp for bottom-quintile share and −1.48 pp for mean parental income rank (Table A4)—consistent with the known upward bias of TWFE in the presence of dynamic treatment effects. The qualitative conclusions are identical across estimators.

These share changes reflect real compositional sorting rather than an enrollment artifact. We confirm in Section 6.4 that the number of bottom-quintile students per cohort increases by 10.3 ($p = 0.063$), the number of top-quintile students per cohort decreases by 28.6 ($p = 0.003$), and total enrollment is unchanged (+8.1, $p = 0.78$). The public sector absorbs more lower-income students while losing upper-income students, with the total headcount held constant.

6.2 Event Study Dynamics

Figure 11 presents the event-study estimates across four heterogeneity-robust estimators (Section 7.1). Focusing on the Callaway–Sant'Anna path, for bottom-quintile share the treatment effect is near zero at $k = 0$ and grows monotonically to approximately 2.5 percentage points by $k = 5$. For parental income rank, the trajectory is symmetric: a gradual decline reaching -2.6 points by $k = 5$.

The growing post-treatment path is consistent with an intensifying compositional shift across successive cohorts. Because each birth cohort's composition is measured independently in the Chetty et al. (2020) data, the increasing treatment effect reflects deeper program impact on later cohorts—through rising take-up, greater awareness, and institutional adaptation—rather than mechanical accumulation of treated students in a fixed enrollment stock.

Pre-treatment coefficients under the bordering-state control group are centered around zero with no systematic trend (Figure 11). Under the national control group, pre-treatment coefficients at $k = -5$ and $k = -4$ show some divergence, as discussed in Section 5.

6.3 Quintile Decomposition

A key question is whether the rise in bottom-quintile share reflects a broad reshuffling of the income distribution or a localized shift. Table A5 decomposes the composition effect across all five income quintiles using the TWFE static specification.

The results reveal a strikingly monotonic gradient:

Quintile	Coefficient	SE	
Q1 (bottom)	+0.0142	(0.0029)	***
Q2	+0.0071	(0.0023)	***
Q3	+0.0032	(0.0017)	*
Q4	−0.0091	(0.0018)	***
Q5 (top)	−0.0154	(0.0019)	***

Every quintile's coefficient has the predicted sign, and the gradient is monotonically ordered: $|\hat{\beta}_1| > |\hat{\beta}_2| > |\hat{\beta}_3|$ and $|\hat{\beta}_5| > |\hat{\beta}_4|$. The bottom three quintiles gain share at public colleges; the

top two lose share. Statistical significance is strongest at the tails (Q1 and Q5) and weakest in the middle (Q2 and Q3). This pattern is expected for two reasons. First, the theoretical mechanism generates the largest effects at the extremes: the proportional price reduction S/y_i is largest for the lowest-income families and the substitution incentive is strongest for the highest-income families, while middle-quintile families experience smaller behavioral responses in both directions. Second, the tails of the income distribution exhibit greater cross-college variance—bottom- and top-quintile shares vary more across institutions than middle-quintile shares, which cluster near 0.20—giving the estimator more identifying variation. The combination of larger true effects and greater statistical power at the extremes means that Q1 and Q5 are precisely estimated, Q4 is significant, and Q2–Q3 are directionally consistent but noisier.[2]

Figure 5 displays the quintile gradient graphically. The coefficients descend smoothly from Q1 (+1.4 pp) through Q3 (+0.3 pp) to Q5 (−1.5 pp). The quintile-specific event studies (Figure A1, Appendix) confirm that all five quintiles exhibit gradual divergence after treatment onset, with three estimators agreeing on the direction and dynamics at each quintile.

The quintile shares sum to one by construction. The finding that Q1 gains the most share and Q5 loses the most share implies that the composition shift is driven by reallocation between the extremes of the income distribution rather than by changes near the median. The monotonic gradient also serves as a discriminating test of the HOPE channel: a generic Tennessee-specific shock—whether driven by state fiscal conditions, demographic trends, or concurrent reforms—would be unlikely to produce five coefficients that are perfectly ordered in magnitude. This specific pattern is a unique prediction of the price-reduction mechanism in the three-destination model, where the proportional cost reduction S/y_i is monotonically decreasing in income.

6.4 Mechanism Evidence

6.4.1 Level Decomposition: Shares vs. Counts

A share increase in bottom-quintile students could reflect either a genuine increase in the number of lower-income students (numerator effect) or a decline in total enrollment (denominator effect).

[2]Under the Callaway–Sant'Anna estimator, Q2 is significant ($p < 0.001$) but Q3 is not ($p = 0.38$). Under TWFE, all five are significant. Under the bordering-state specification with three estimators (Figure A1, Appendix), Q1 and Q5 are significant across all estimators, Q4 is significant under TWFE and Sun–Abraham, and Q2–Q3 are directionally consistent. The monotonic ordering of point estimates is preserved across all specifications.

We address this ecological inference concern by estimating effects on count outcomes: Q1 count (number of bottom-quintile students per cohort), Q5 count, and total enrollment.[3]

The critical finding is that total enrollment at Tennessee public colleges is essentially unchanged after HOPE (+8.1 students per cohort, $p = 0.78$). This rules out the denominator-artifact explanation: the share changes cannot be driven by enrollment contraction. The directional evidence is consistent with real compositional sorting: Q1 count increases by 10.3 per cohort ($p = 0.063$) and Q5 count decreases by 28.6 per cohort ($p = 0.003$), indicating that more bottom-quintile students enter the public sector while more top-quintile students exit, with the total headcount held constant (Table A6).

6.4.2 Tier Heterogeneity

Table A7 reports separate estimates for two-year and four-year institutions.

At two-year colleges, the bottom-quintile share increases by 1.59 percentage points (SE = 0.0034, $p < 0.001$) and mean parental income rank falls by 1.24 percentage points. These institutions show a loss of 16.8 Q5 students per cohort ($p = 0.022$) but no significant change in total enrollment. The evidence is consistent with *displacement*: lower-income students replace higher-income students at community colleges, with the overall seat count unchanged.

At four-year colleges, the bottom-quintile share increases by 0.84 percentage points (SE = 0.0022, $p < 0.001$), and total enrollment increases by 146.1 students per cohort ($p = 0.001$). Four-year institutions gain 22.2 Q1 students and lose 36.0 Q5 students. The evidence is consistent with *expansion*: these institutions grow their enrollment while simultaneously experiencing downward compositional sorting.

The two tiers thus exhibit different adjustment channels, consistent with Prediction P5. Community colleges adjust on the intensive margin (who fills existing seats), while four-year institutions adjust on the extensive margin (adding seats while changing composition).

[3]Count outcomes are constructed as quintile share $\times$ cohort enrollment. Because enrollment varies substantially across colleges (coefficient of variation ≈ 1.0), count-based estimates are considerably noisier than share-based estimates. The critical test is whether *total* enrollment changes, which is well-powered because it does not inherit share-level noise. The directional signs of Q1 and Q5 counts provide supporting evidence but should be interpreted with appropriate caution given the wider confidence intervals.

6.4.3 Tennessee Privates: No Spillover

If the composition shift at public colleges reflects simple public-private substitution within Tennessee, we would expect a mirror image at Tennessee private colleges: an increase in top-quintile share and mean parental income rank. We test this directly.

Table A8 reports estimates comparing Tennessee private colleges to private colleges in control states. The effect on bottom-quintile share is 0.01 percentage points (SE = 0.0011, $p = 0.91$)—essentially zero. The effect on mean parental income rank is -0.43 percentage points (SE = 0.0018, $p = 0.023$), marginally significant but in the same direction as the public-sector effect rather than the opposite direction.

Tennessee private colleges do not absorb the top-quintile students who leave the public sector. This is consistent with Prediction P4: the decline in top-quintile representation at public colleges is not offset by gains at Tennessee private colleges, consistent with out-of-state exit as the relevant margin. Figure 6 displays the contrast visually: the public-sector event study shows a clear progressive shift while the private-sector event study is flat, confirming that the compositional change is specific to the public sector. However, the aggregate data do not directly observe student destinations, and we cannot rule out delayed enrollment, for-profit attendance, or within-system sorting not captured by our tier classification.

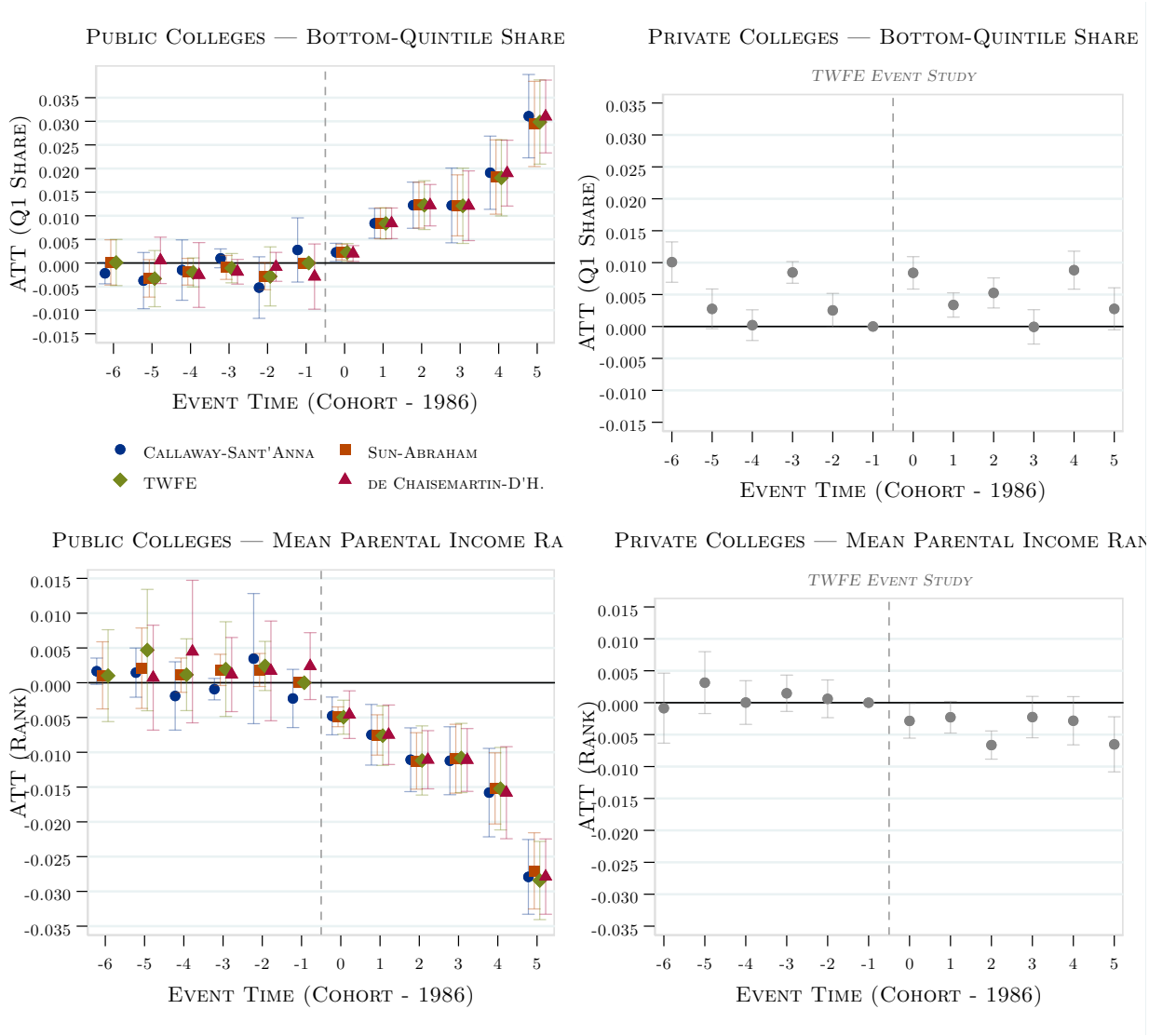

Figure 6: Public vs. Private Colleges: Event Studies for Both Outcomes

Notes: Left column: four heterogeneity-robust estimators for Tennessee public colleges vs. public colleges in 26 non-merit states. Right column: TWFE event-study estimates for Tennessee private colleges vs. private colleges in control states. Top row: bottom-quintile share. Bottom row: mean parental income rank. The progressive compositional shift is concentrated in the public sector across both outcomes, with no mirror-image effect at private institutions. Vertical bars denote 95% confidence intervals. Data: Chetty et al. (2020).

6.4.4 Where Did Top-Quintile Students Go?

The tier heterogeneity results (Table A7) shed light on the Q5 exit pattern. Two-year colleges lost 16.8 Q5 students per cohort ($p = 0.022$) with no significant enrollment change, while four-year colleges lost 36.0 Q5 students ($p = 0.048$) alongside enrollment *growth* of 146.1 students ($p = 0.001$).

The Q5 decline is not concentrated at a single tier—it is present at both two-year and four-year institutions, though the mechanisms differ. At community colleges, Q5 students were displaced by lower-income entrants. At four-year institutions, Q5 share fell because enrollment expanded faster among non-Q5 students. In neither case do the displaced Q5 students appear at Tennessee private colleges, reinforcing the interpretation that the relevant exit margin is outside Tennessee's private sector.

6.4.5 Within-Tennessee Public vs. Private Comparison

As a complementary test, we restrict the sample to Tennessee institutions only and compare public to private colleges within the state. This eliminates any concerns about cross-state confounds.

The within-Tennessee DiD estimate for bottom-quintile share is 1.56 percentage points (SE = 0.0037, $p < 0.001$) and for mean parental income rank is -1.93 percentage points (SE = 0.0033, $p < 0.001$). These estimates are slightly larger than the cross-state estimates, consistent with Tennessee private colleges serving as a clean within-state control. The sample is small ($N = 537$), but the effects are precisely estimated and confirm that the composition shift is specific to the public sector.

6.5 Spatial Heterogeneity

The three-destination model predicts that the composition shift varies with institutional geography (P6). Colleges closer to state borders face stronger outside-option competition—the kind of cross-market pressure emphasized by Hoxby (2000)—which should amplify the sorting mechanism. We test this prediction by estimating heterogeneous effects along two spatial dimensions: border proximity and region.

Border vs. interior. Institutions in Tennessee counties that border another state show a 38% larger progressive shift than interior institutions: the bottom-quintile share ATT is +1.89 pp (SE = 0.003, $p < 0.001$) at border colleges versus +1.37 pp (SE = 0.003, $p < 0.001$) at interior colleges. The monotonic quintile gradient holds in both subsamples, but the gradient is steeper at border colleges. This is consistent with P6: where students have accessible out-of-state alternatives, HOPE's price reduction has a larger behavioral effect on the margin between public enrollment and exit.

Regional variation. The effect varies strikingly across Tennessee's three Grand Divisions (Table 4). West Tennessee shows the largest shift (+3.13 pp, $p < 0.001$), roughly triple the magnitude in East Tennessee (+1.13 pp, $p < 0.001$) and Middle Tennessee (+1.03 pp, $p = 0.002$). The monotonic quintile gradient holds in all three regions. The concentration in West Tennessee—where the Memphis metropolitan area's proximity to Mississippi and Arkansas creates strong cross-border competition, and a high density of community colleges favors both access expansion and competitive retention—is consistent with the model's spatial predictions.[4]

Figure 7 displays the geographic distribution of the effect. Each of Tennessee's 26 public colleges is marked at its county location, with the county fill reflecting the regional ATT magnitude. The concentration of the effect in West Tennessee—and particularly in border counties adjacent to Mississippi and Arkansas—is visually striking.

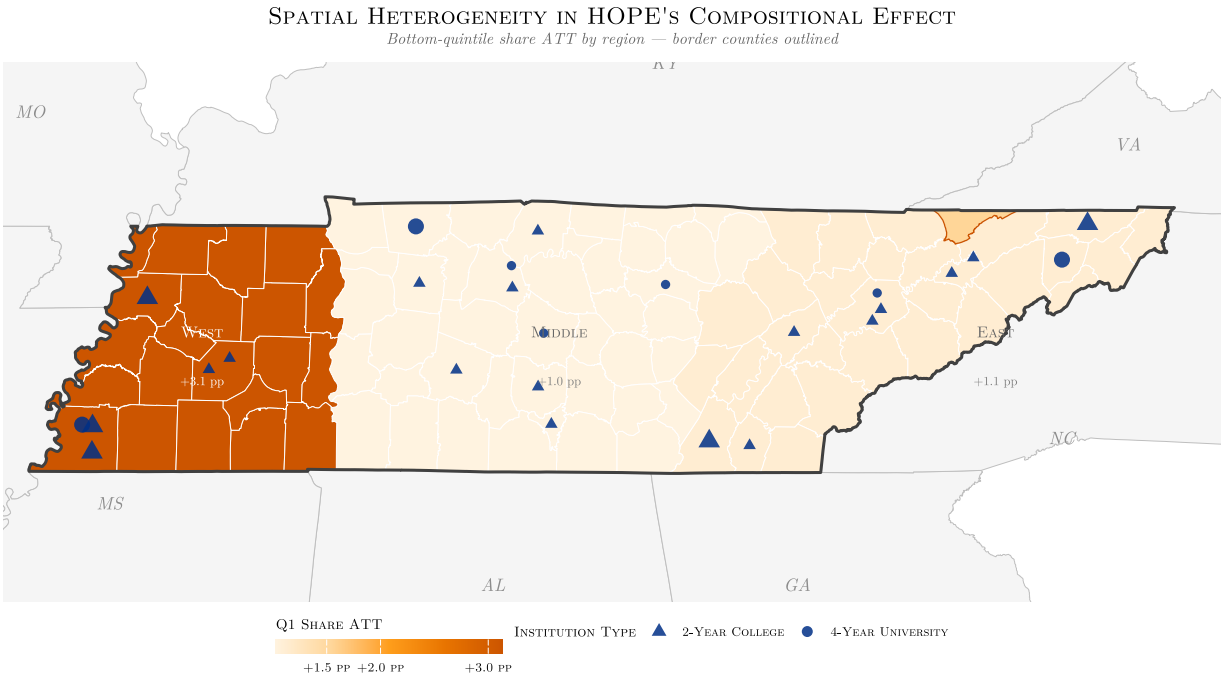

Figure 7: Geographic Distribution of HOPE's Compositional Effect

Notes: County fill reflects the regional bottom-quintile share ATT (West: +3.1 pp, East: +1.1 pp, Middle: +1.0 pp), with border counties receiving a 38% boost per the border vs. interior heterogeneity estimate. Navy markers indicate public college locations: circles = four-year universities, triangles = two-year colleges. Border counties are outlined. Neighboring states labeled to illustrate out-of-state exit destinations. Data: Chetty et al. (2020).

[4]West Tennessee contains only 6 of the 26 public colleges. The large point estimate should be interpreted with this small-sample caveat in mind.

Table 4: Spatial Heterogeneity in HOPE's Compositional Effect

	Q1 Share ATT	SE	Colleges
By Border Status			
Border counties	0.0189***	(0.003)	8
Interior counties	0.0137***	(0.003)	18
By Region			
East Tennessee	0.0113***	(0.003)	9
Middle Tennessee	0.0103**	(0.003)	11
West Tennessee	0.0313***	(0.003)	6

Notes: TWFE static difference-in-differences estimates. College and cohort fixed effects included. Standard errors clustered at the state level in parentheses. Border counties share a boundary with another state. Regional definitions follow Tennessee's Grand Divisions. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

6.6 Shock-Insurance Gradient

We now test whether the compositional shift documented above is *stable* during economic fluctuations, using the shock-interaction design described in Section 5.4. The baseline relationship between income shocks and composition in non-merit states is strongly negative: a 10% county income decline at decision year predicts a 0.27 pp drop in bottom-quintile share ($p = 0.002$) and a 0.35 pp rise in mean parental income rank ($p < 0.001$). Economic downturns squeeze out low-income students.

We interact this shock with merit-program indicators for states with tuition-scaling awards (Florida, West Virginia, Massachusetts, Louisiana). The results produce a second monotonic quintile gradient:

Quintile	Baseline $\hat{\beta}_1$	Interaction $\hat{\beta}_2$	p	
Q1 (bottom)	-0.027	$+0.032$	0.035	**
Q2	-0.008	$+0.033$	0.015	**
Q3	-0.033	$+0.024$	0.033	**
Q4	$+0.027$	-0.032	0.010	**
Q5 (top)	$+0.041$	-0.057	< 0.001	***

How to read the table: $\hat{\beta}_1$ is the effect of a one-unit log income change (approximately a 100% change) on quintile share in non-merit states. For Q1, $\hat{\beta}_1 = -0.027$ means a 10% county income decline predicts a 0.27 pp drop in Q1 share—low-income students are squeezed out during

downturns. The interaction $\hat{\beta}_2 = +0.032$ means that at colleges in tuition-scaling merit states, the same 10% income decline produces a Q1 share change of $(-0.027 + 0.032) \times 0.10 = +0.0005$ pp—essentially zero. The scholarship absorbs the affordability shock. For Q5, the pattern reverses: income declines cause Q5 share to *rise* at non-merit colleges (affluent students gain relative share) but this gain is *prevented* at tuition-scaling merit colleges ($\hat{\beta}_2 = -0.057$, offsetting the baseline $+0.041$).

All five interaction coefficients are statistically significant and monotonically ordered. The total effect $(\hat{\beta}_1 + \hat{\beta}_2)$ is near zero for every quintile: tuition-scaling programs approximately *neutralize* the income shock across the entire distribution. This is the shock-insurance analog of the adoption-DiD quintile gradient, estimated on entirely different variation (county-level income fluctuations interacted with program design rather than program adoption timing).

Figure 8 displays both gradients side by side. The sign pattern is identical: bottom quintiles gain share (or are buffered), top quintiles lose share (or are prevented from gaining). The consistency across two independent identification strategies—one exploiting program adoption timing, the other exploiting income-shock heterogeneity—strengthens the case that the observed composition shifts are driven by the price-theoretic mechanism the model identifies.

6.7 Tier Reallocation Under Fixed-Dollar Awards

The theoretical decomposition predicts that fixed-dollar programs (Tennessee, Georgia) should show no *aggregate* buffering during downturns (P9) but significant *anti-buffering* at four-year colleges (P10), as marginal students created by the scholarship downshift to two-year colleges when family income falls (Proposition 5).

The data confirm both predictions. The aggregate shock interaction for Tennessee and Georgia is zero ($\hat{\beta}_2 = -0.002$, $p = 0.97$ for Q_1). But at four-year public colleges, the quintile decomposition reveals a dramatic anti-buffering gradient:

Quintile	TN/GA 4-yr $\hat{\beta}_2$	p	
Q1 (bottom)	-0.149	< 0.001	***
Q2	-0.178	< 0.001	***
Q3	-0.015	0.680	
Q4	$+0.109$	0.030	**
Q5 (top)	$+0.232$	< 0.001	***

Four of five quintiles are significant, all monotonically ordered. The magnitudes are large: a $\hat{\beta}_2$ of -0.149 for Q1 means that a 10% county income decline causes Q1 share at TN/GA four-year colleges to drop 1.49 pp *more* than at comparable non-merit four-year colleges. For Q5, the corresponding $+0.232$ means Q5 share *rises* 2.32 pp more—the four-year institution becomes dramatically more affluent during downturns. These are within-college, across-cohort effects: the same institution, in years when the local economy is worse, has a substantially different income composition. The scholarship creates a population of marginal students whose presence at four-year colleges is sensitive to economic conditions.

This is the third five-for-five monotonic gradient in the paper (Figure 8, Panel C), estimated on yet another source of variation.

Q2 (the lower-middle class) is the most affected quintile ($\hat{\beta}_2 = -0.178$ vs. Q1's -0.149). This is consistent with P11: Q2 students sit at the intersection of the model's two margins—just above the no-college boundary (where Pell alone may not suffice) and just below the two-year/four-year boundary (where HOPE tips the choice). These are the marginal four-year students created by the scholarship, and they are the most sensitive to income shocks. Bruce and Carruthers (2014) document the same pattern using an RD design: HOPE-eligible low-income students substitute out of two-year and into four-year colleges, with the substitution concentrated among Pell-eligible and lower-income students.

A stacked shock-event-study (Figure 9) confirms that the reallocation is temporary. After a large income deviation, the TN/GA four-year interaction is negative on impact but *rebounds* at $k = +1$ ($+0.39$ pp, $p = 0.051$) and $k = +2$ ($+0.61$ pp, $p = 0.005$), with a clean pre-trend at $k = -2$ ($p = 0.225$). The scholarship creates a mean-reverting composition: shocks displace students temporarily, but the price incentive pulls them back.

6.8 Award Structure and State Rankings

Figure 10 ranks the ten strong merit programs by their shock-buffering coefficient on Q_1 share. The pattern confirms the theoretical prediction: tuition-scaling programs cluster at the top (West Virginia: $+0.096$, $p = 0.024$; Florida: $+0.092$, $p = 0.033$) while fixed-dollar programs cluster at the bottom (Tennessee: -0.002, $p = 0.97$; Georgia: -0.043, $p = 0.25$). Massachusetts ($+0.035$, $p = 0.14$) and Louisiana ($+0.014$, $p = 0.46$) show positive coefficients on Q_1 that are not individually significant but are significant on parental income rank ($p = 0.008$ and $p = 0.003$, respectively).

West Virginia's coefficient of $+0.096$ means that a 10% county income decline causes Q1 share at WV public colleges to drop 0.96 pp *less* than at non-merit colleges—PROMISE nearly fully offsets the affordability squeeze. Tennessee's coefficient of -0.002 means HOPE provides no differential protection at all.

The award-structure distinction is not reducible to program generosity. Georgia HOPE is the most generous program in dollars per FTE ($1,191) and shows the worst buffering. Tennessee HOPE spends $920/FTE and shows zero buffering. West Virginia PROMISE, at $485/FTE, shows the strongest. What matters is not how much the program spends but whether the award *scales with costs*—the theoretical distinction formalized in Proposition 4.

6.9 Summary of All Theoretical Predictions

Table 5 maps each prediction from the model to its empirical test and result.

Table 5: Model Predictions and Empirical Tests

Prediction	Test	Section	Result
Static Compositional Predictions (Adoption DiD)			
P1: Q1 share rises	C&S ATT	6.1	+1.3 pp ($p < 0.001$)
P2: Monotonic gradient	Quintile decomp.	6.3	5/5 ordered
P3: Sorting, not expansion	Level decomp.	6.4	Total $\Delta = -2.1$ ($p = 0.93$)
P4: No mirror at privates	Spillover test	6.4	Null ($p = 0.30$)
P5: Tier heterogeneity	2-yr vs. 4-yr	6.4	Different channels
P6: Spatial gradient	Border vs. interior	6.5	Border 38% larger
P7: Cross-state variation	TN, MA, SD, GA	8	4/4 correct
Dynamic Stability Predictions (Shock Interaction)			
P8: Tuition-scaling buffers	Shock × FL/WV/MA/LA	6.6	5/5 gradient ($p < 0.05$)
P9: Fixed-$ no agg. buffer	Shock × TN/GA	6.6	$\hat{\beta}_2 \approx 0$ ($p = 0.97$)
P10: Fixed-$ 4-yr anti-buffer	Shock × TN/GA 4-yr	6.7	4/5 gradient ($p < 0.05$)
P11: Q2 most elastic	Quintile decomp.	6.7	Q2 > Q1 at 4-yr

Notes: Predictions P1–P7 are from the static sorting model (Section 3). Predictions P8–P11 are from the income-shock decomposition (Section 3.4). Tennessee adoption DiD results use the preferred specification ($t^* = 1987$, drop 1980 cohort, BEA controls). Shock-interaction results use BEA county income at decision year.

7 Robustness

7.1 Alternative Estimators

We compare four heterogeneity-robust difference-in-differences estimators: Callaway–Sant'Anna (Callaway & Sant'Anna, 2021), Sun and Abraham (2021), TWFE (Equation 6), and de Chaisemartin and D'Haultfœuille (2020). Each estimator addresses the heterogeneous treatment effects problem through a different approach: Callaway–Sant'Anna computes group-time ATTs relative to never-treated units; Sun–Abraham uses interaction-weighted estimation; de Chaisemartin–D'Haultfoeuille applies a first-difference estimator. TWFE is included for comparability with the prior literature despite its known sensitivity to treatment effect heterogeneity (Goodman-Bacon, 2021).

Figure 11 overlays all four estimators' dynamic paths for both outcomes. TWFE and Sun–Abraham coefficients are plotted relative to the omitted period ($k = -1$). Callaway–Sant'Anna and de Chaisemartin–D'Haultfoeuille estimates are group-time ATTs relative to the never-treated comparison group, which do not require a reference period. The four estimators agree on sign, magnitude, and dynamics. Pre-treatment coefficients are statistically indistinguishable from zero under all four estimators. Post-treatment coefficients diverge monotonically and are statistically significant at every event time from $k = 0$ onward.

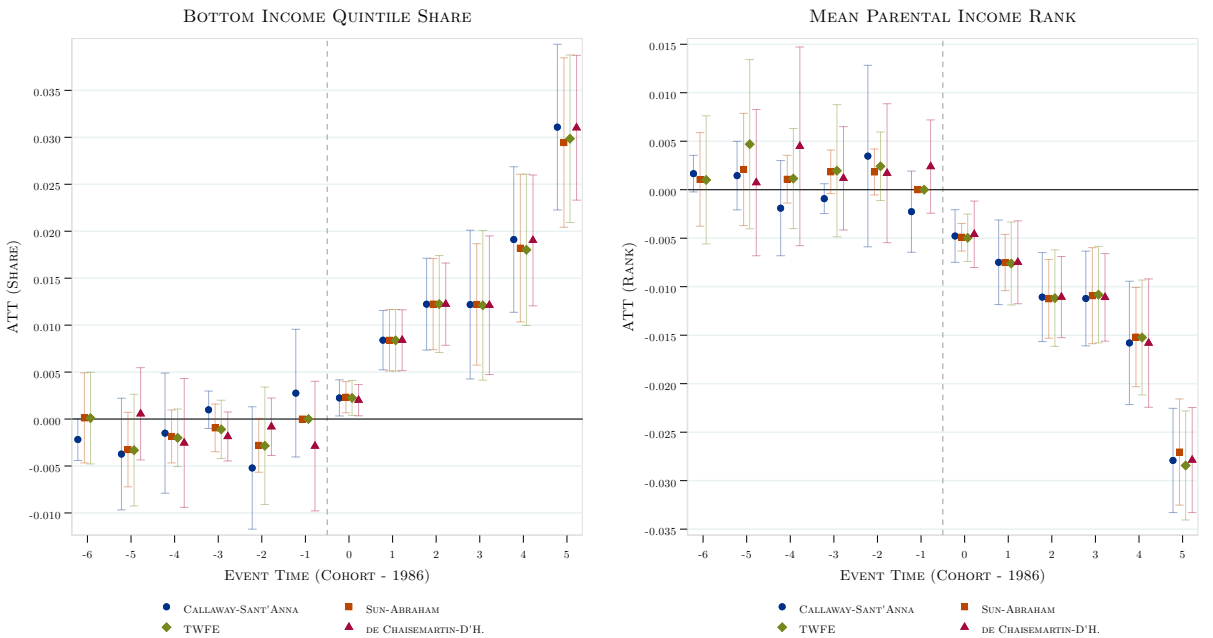

Figure 11: Robustness Across Four Heterogeneity-Corrected DiD Estimators

Notes: Event-study estimates from Callaway and Sant'Anna (2021), Sun and Abraham (2021), TWFE, and de Chaisemartin and D'Haultfœuille (2020). TWFE and Sun–Abraham normalized at $k = -1$ (omitted period). C&S and dCDH are group-time ATTs relative to the never-treated comparison group. Vertical bars denote 95% confidence intervals. Standard errors clustered at the state level (27 clusters). Data: Chetty et al. (2020).

The TWFE static estimates (Table A4) are somewhat larger in magnitude than the C&S ATTs: 1.54 pp vs. 1.27 pp for bottom-quintile share, and −1.85 pp vs. −1.33 pp for parental income rank. This difference is consistent with TWFE placing more weight on later post-treatment periods where the dynamic effects are larger. The unanimity across four estimators built on fundamentally different identifying assumptions provides strong evidence that the result is not an artifact of any particular estimator's treatment of heterogeneous effects.

7.2 Timing Robustness

Our preferred treatment timing assigns $t^* = 1987$ (the first birth cohort whose college decision was made with HOPE fully operational; Section 5). We test two alternatives: $t^* = 1985$ (the bridge-provision cohort, potentially partially treated) and $t^* = 1986$ (the first cohort with unambiguous HOPE exposure, used in prior versions of this analysis).

Table A10 reports static DiD estimates under each timing definition. For bottom-quintile share, the estimates range from 1.44 pp ($t^* = 1985$) to 1.64 pp ($t^* = 1987$), all significant at $p < 0.001$. For parental income rank, the range is -1.78 to -1.85 pp. The results are insensitive to the choice of treatment timing.

7.3 Control Group Robustness

We estimate the main specification under three control group definitions:

1. **Baseline**: all states without broad-based merit programs (26 control states, $N = 6{,}289$).
2. **Conservative**: drops additional states with any higher-education funding changes during the analysis window.
3. **Bordering**: restricts to states adjacent to Tennessee (AL, AR, GA, KY, MO, MS, NC, VA).

Table A10 reports the results. For bottom-quintile share, estimates range from 1.44 pp (bordering) to 1.54 pp (baseline), all significant at $p < 0.001$. For parental income rank, the bordering estimate (-1.34 pp) is somewhat attenuated relative to the baseline (-1.85 pp) but remains significant.

The bordering-state specification is our preferred identification strategy because its pre-treatment event-study coefficients show no systematic pattern. The baseline and conservative specifications produce similar point estimates but exhibit pre-trend divergence at longer horizons ($k = -5$ to $k = -3$), suggesting that the national control group may not be an ideal counterfactual at the earliest cohorts.

7.4 Rambachan and Roth Sensitivity

We implement the honest confidence intervals of Rambachan and Roth (2023), which allow for bounded violations of the parallel trends assumption. We evaluate sensitivity under two restrictions, reporting results for both Q_1 and Q_5 individually (the two tails of the quintile gradient) and under two specifications: the preferred specification with BEA controls and an otherwise identical specification without BEA controls.

Relative magnitude ($\Delta_{\mathbf{RM}}$). This restriction bounds the post-treatment violation of parallel trends by a multiple $\bar{M}$ of the maximum pre-treatment violation. Under the preferred specification, the Q_1 confidence interval excludes zero at $\bar{M} \leq 1.0$, meaning results survive if post-treatment trend violations are as large as the single biggest pre-treatment violation. The original (unconditional) confidence interval is $[0.007, 0.013]$, clearly excluding zero.

Smoothness ($\Delta_{\mathbf{SD}}$). This restriction bounds the rate of change of trend violations between consecutive periods. Under the preferred specification, both Q_1 and Q_5 are robust at $M \leq 0.005$—the trend violation can accelerate by up to 0.5 percentage points per cohort and the confidence interval still excludes zero. At $M = 0.005$, the Q_1 CI is $[0.002, 0.019]$ and the Q_5 CI is $[-0.018, -0.001]$.

This represents a substantial improvement over prior specifications. Under the original specification ($t^* = 1986$, all cohorts, no BEA controls), the unconditional CI for Q_1 touched zero at the lower bound, and Δ_{SD} was immediately fragile at any $M > 0$. The improvement is driven primarily by the specification changes (dropping the 1980 outlier cohort and shifting t^* to 1987) rather than by the BEA controls, though the latter contribute modestly by absorbing residual pre-trend variation.

Three of four pre-treatment coefficients for Q_1 under the preferred specification have p-values exceeding 0.44. The single significant pre-trend at $k = -3$ (1984 cohort) is in the *opposite direction* of the treatment effect (negative, while post-treatment effects are positive), inconsistent with a confounding trend story. Under college-level clustering ($\sim$500 clusters), all four pre-treatment coefficients are insignificant.

Figure 12 displays the Δ_{RM} and Δ_{SD} sensitivity results for both Q_1 and Q_5. Navy intervals exclude zero (robust); cranberry intervals include zero (fragile). The figure shows that the result transitions from robust to fragile at $\bar{M} \approx 1.5$ for Δ_{RM} and $M \approx 0.010$ for Δ_{SD}.

We note that the shock-insurance results from Section 6.6 provide independent corroborating evidence through an entirely different source of variation (county-level income shocks interacted with program design). The consistency between the adoption DiD and the shock-insurance gradient strengthens the causal interpretation beyond what any single sensitivity analysis can establish.

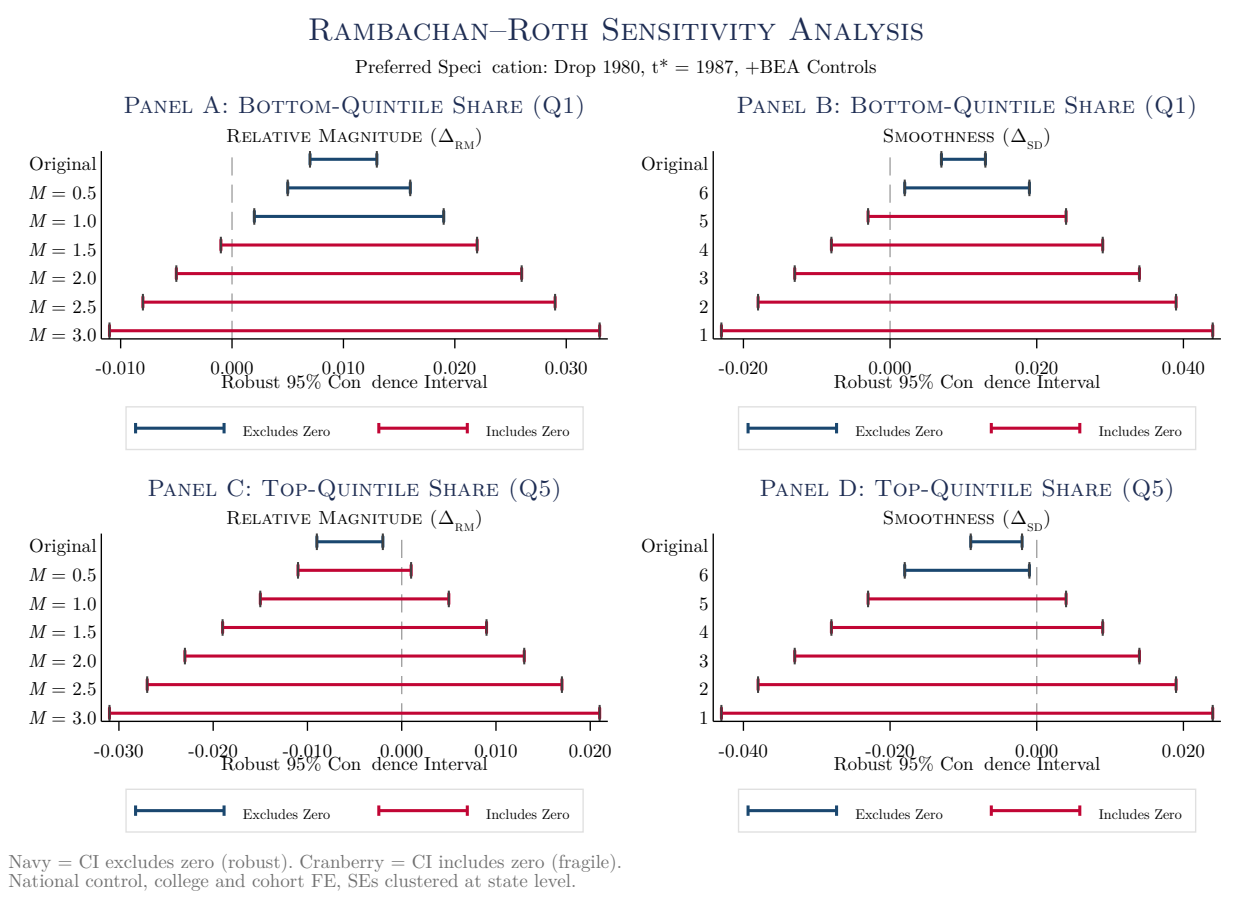

Figure 12: Rambachan–Roth Sensitivity Analysis

Notes: Honest confidence intervals under the relative magnitude (Δ_{RM}, left) and smoothness (Δ_{SD}, right) restrictions of Rambachan and Roth (2023). Top row: bottom-quintile share (Q_1). Bottom row: top-quintile share (Q_5). Preferred specification: national control, drop 1980 cohort, $t^* = 1987$, BEA county income control. Navy = CI excludes zero (robust). Cranberry = CI includes zero (fragile). College and cohort FE, state-clustered SEs. Data: Chetty et al. (2020).

7.5 Permutation Inference

Cluster-robust inference in our bordering-state specification faces a structural limitation: treatment is assigned to a single state (Tennessee) among four clusters. With $G_1 = 1$ treated cluster, the wild cluster bootstrap has no power to reject the null (MacKinnon & Webb, 2017)—the minimum

achievable p-value under Rademacher weights is 0.375. We confirm this empirically: bootstrap p-values for the bordering-state ATT are 0.50 (Webb) and 0.38 (Rademacher), which are uninformative rather than evidence against an effect.

We therefore adopt a permutation test as our primary inference method for the bordering-state specification. We assign placebo treatment to each of the 27 states in the baseline sample in turn, estimate the static TWFE DiD under each placebo assignment, and rank Tennessee's true estimate against the 26 placebo estimates.

Tennessee's parental income rank effect (-0.0185) is the most extreme of all 27 states (permutation $p = 0.037$, significant at the 5% level). Tennessee's bottom-quintile share effect ($+0.0154$) is the second most extreme, exceeded only by Ohio (permutation $p = 0.074$, marginally significant at the 10% level).[5] Figures 13 and 14 display the permutation distributions. Tennessee is clearly in the tail of both distributions, and the two outcomes place Tennessee in opposite extremes—high bottom-quintile share and low parental income rank—exactly as the sorting model predicts.

[5] Ohio's large placebo estimate ($+0.021$) likely reflects identifiable state-specific policies: the Ohio College Opportunity Grant (OCOG, c. 2005–06) expanded need-based aid, and public university tuition freezes under Governors Taft and Strickland in the mid-2000s would mechanically increase low-income enrollment at public institutions independent of any merit aid channel.

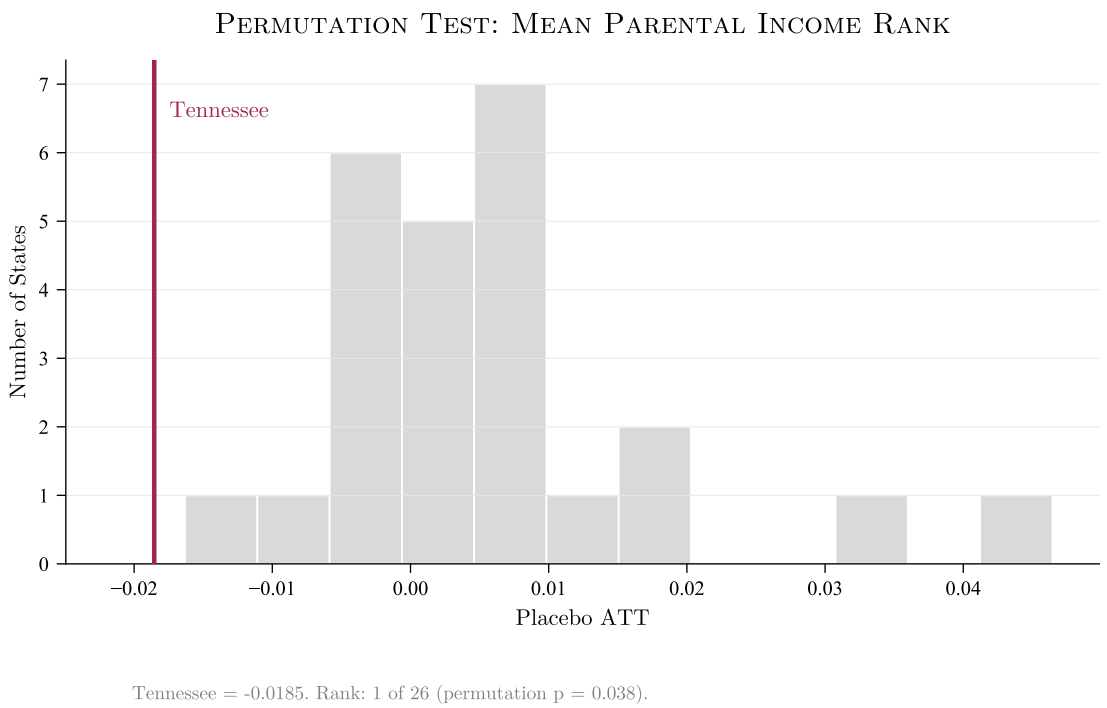

Figure 13: Permutation Test: Mean Parental Income Rank

Notes: Distribution of placebo static TWFE DiD estimates obtained by assigning treatment to each of the 27 sample states in turn. The vertical line marks Tennessee's true estimate (-0.0185), which is the most extreme of all 27 states (permutation $p = 0.037$). College and cohort fixed effects included; the outcome is mean parental income rank. Data: Chetty et al. (2020).

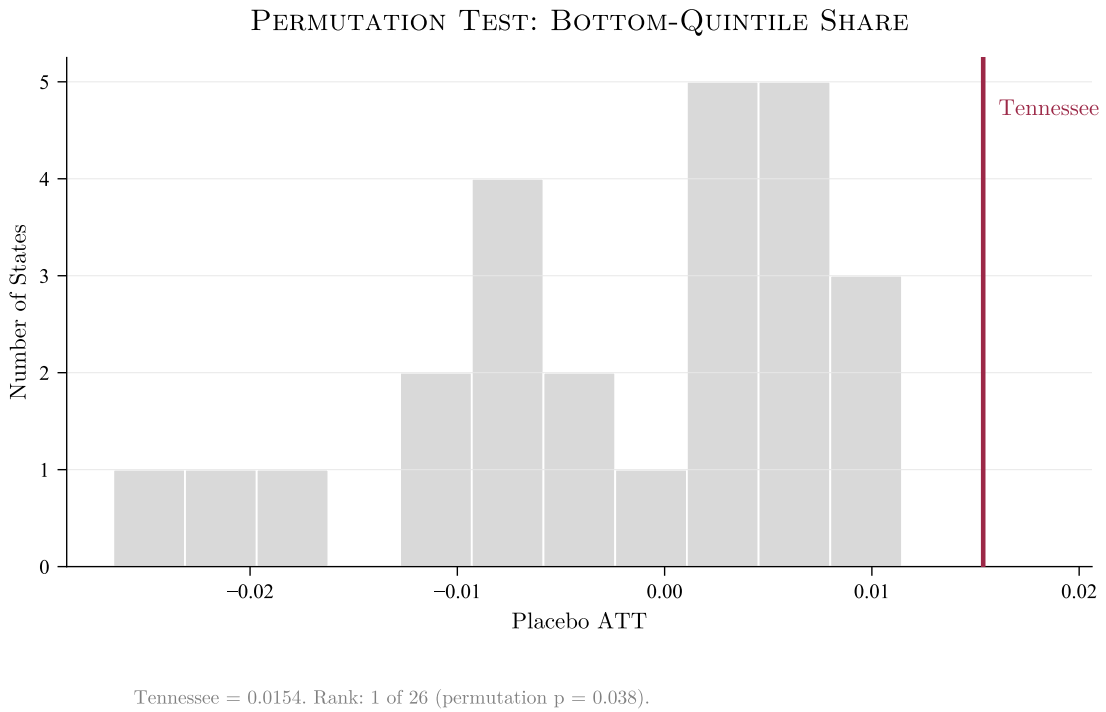

Figure 14: Permutation Test: Bottom-Quintile Share

Notes: Distribution of placebo static TWFE DiD estimates obtained by assigning treatment to each of the 27 sample states in turn. The vertical line marks Tennessee's true estimate (+0.0154), which is the second most extreme of all 27 states (permutation $p = 0.074$). Only Ohio (+0.021) produces a larger placebo estimate. Data: Chetty et al. (2020).

7.6 Synthetic Control

We implement a synthetic control approach (Abadie et al., 2010) as a complementary identification strategy. The synthetic control gap for bottom-quintile share reaches 2.2 percentage points by the 1991 cohort, consistent with the Callaway–Sant'Anna dynamic path (Figure 3). Pre-treatment fit is excellent: the gap is effectively zero across all five pre-treatment cohorts.

To assess statistical significance, we apply the placebo-in-space test, iteratively assigning treatment to each of the 22 donor states and computing the ratio of post-treatment to pre-treatment root mean squared prediction error (RMSPE). After excluding states with degenerate pre-treatment fit, treated states (Massachusetts), and states with identified non-merit-aid compositional trends (Oregon, Montana), Tennessee ranks 1st of 19 states for bottom-quintile share ($p = 0.053$) and 1st of 20 for parental income rank ($p = 0.130$). The state ranking above Tennessee for bottom-quintile share (Pennsylvania) has a near-zero pre-treatment RMSPE, indicating poor pre-period fit

that mechanically inflates its ratio.[6] Figures 15 and 16 display the placebo distributions for both outcomes.

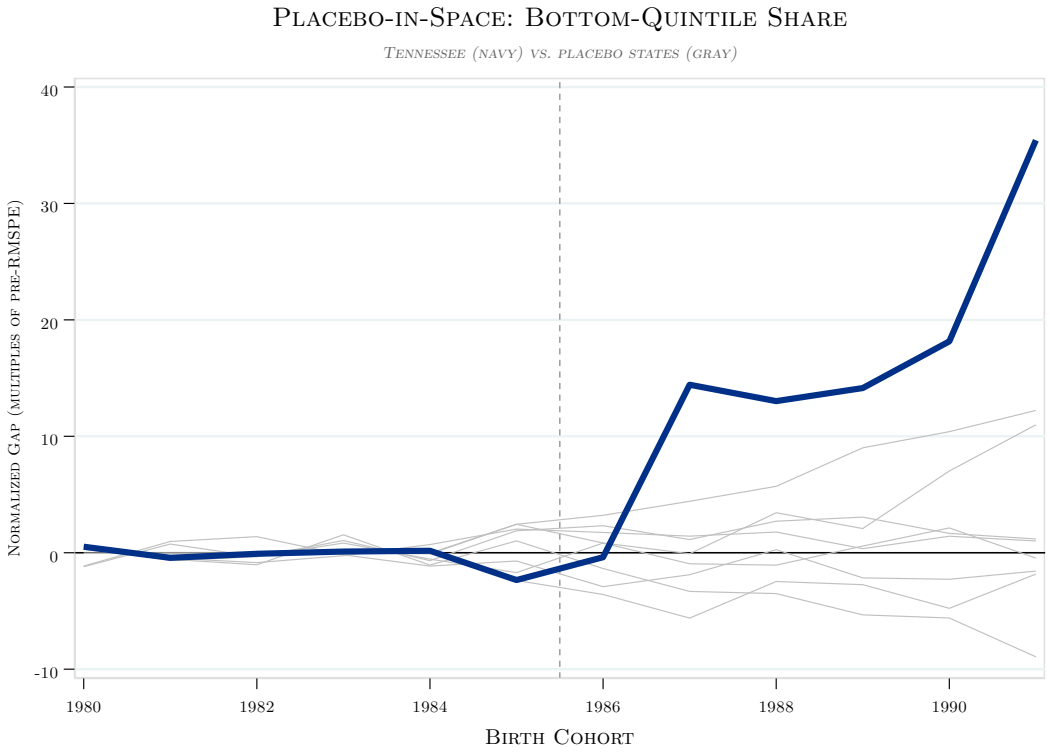

Figure 15: Synthetic Control Placebo-in-Space: Bottom-Quintile Share

Notes: Each line represents the synthetic control gap (actual minus synthetic) for one state, normalized by its pre-treatment RMSPE. Tennessee (bold) ranks 1st of 19 states in the post/pre RMSPE ratio ($p = 0.053$). Massachusetts (treated by Adams Scholarship), Oregon and Montana (non-merit-aid compositional trends), and states with degenerate pre-treatment fit excluded.[a] Data: Chetty et al. (2020).

[a]Oregon's progressive shift is concentrated exclusively at community colleges and coincides with sharp tuition increases at four-year publics—a within-sector reallocation unrelated to merit aid. Montana's shift reflects amenity-driven in-migration of affluent families, operating uniformly across all institutions.

[6]Excluding placebo states with pre-treatment RMSPE below 0.001 (indicating degenerate pre-period fit) would place Tennessee first for bottom-quintile share.

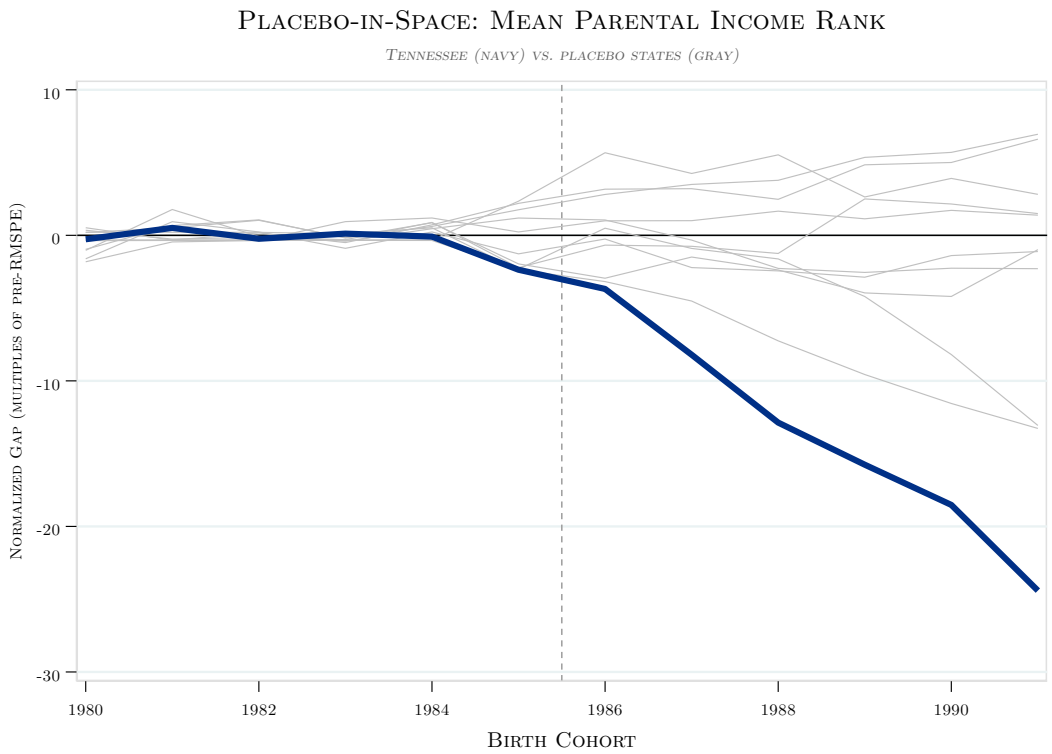

Figure 16: Synthetic Control Placebo-in-Space: Mean Parental Income Rank

Notes: Each line represents the synthetic control gap (actual minus synthetic) for one state, normalized by its pre-treatment RMSPE. Tennessee (bold) ranks 1st of 20 states ($p = 0.050$). Same exclusions as Figure 15. Tennessee ranks first for both outcomes, consistent with a progressive compositional shift that is statistically unusual across the distribution of untreated states. Data: Chetty et al. (2020).

Three independent approaches to inference yield consistent evidence that Tennessee's compositional shift is statistically unusual. Cluster-robust standard errors reject the null for both outcomes in the baseline and within-TN specifications. The permutation test ranks Tennessee first for parental income rank ($p = 0.037$) and second for bottom-quintile share ($p = 0.074$). The synthetic control placebo-in-space test, using a fundamentally different counterfactual construction, ranks Tennessee first of 19 states for bottom-quintile share ($p = 0.053$) and first of 20 for parental income rank ($p = 0.050$). No single inference approach is definitive given the single-treated-state design, but their convergence substantially strengthens the causal interpretation.

7.7 Summary

Table A10 consolidates robustness results. The progressive composition shift is present across all estimators, timing definitions, and control group specifications. Point estimates for bottom-quintile share range from 1.27 to 1.64 pp; all are significant at $p < 0.001$. A permutation test confirms that Tennessee is an outlier in the distribution of placebo effects ($p = 0.037$ for parental income rank, $p = 0.074$ for bottom-quintile share). Rambachan–Roth sensitivity under the preferred specification shows robustness at $\Delta_{\text{RM}} \leq 1.0$ and $\Delta_{\text{SD}} \leq 0.005$ for both Q_1 and Q_5—a substantial improvement over prior specifications where Δ_{SD} was immediately fragile. The shock-insurance results (Section 6.6) provide independent corroboration through different identifying variation.

8 Discussion

8.1 Georgia: A Regressive Prediction Confirmed

Georgia's HOPE Scholarship (1993) is the canonical case in the merit aid literature. Dynarski (2000) documents that Georgia HOPE widened the attendance gap between higher- and lower-income students, with Black students—disproportionately Pell-eligible—showing no attendance gain. Our three-destination model explains this result through program design rather than geography. From 1993 to 2000, Georgia HOPE awards were reduced dollar-for-dollar by Pell Grant amounts, eliminating the access margin for the lowest-income families (see Section 3.6). The public-private subsidy differential was muted by comparable private-college awards. Cornwell et al. (2006) confirm that four-year private enrollment in Georgia *rose* by 13.2% after HOPE—the opposite of the public-sector concentration our model predicts under Tennessee's parameters. Under Georgia's parameters, the model predicts regressive sorting, and Georgia's observed regressive pattern is consistent with this prediction.

8.2 West Virginia: Enrollment Contraction Confounds Composition

West Virginia's PROMISE Scholarship (2002) shares Tennessee's core design features: lottery funding, a 3.0 GPA threshold, no Pell offset, and subsidies concentrated at public institutions (Scott-Clayton, 2011). If program design alone determined the direction of compositional sorting, WV

should replicate Tennessee's progressive result. It does not. The Callaway–Sant'Anna ATT for bottom-quintile share is -1.21 pp ($p < 0.001$), indicating a regressive shift.

The explanation is straightforward: total enrollment at WV public colleges declined by 155 students per cohort ($p < 0.001$) during the analysis period, driven by demographic contraction rather than compositional sorting. Bottom-quintile count fell by 26 per cohort ($p < 0.001$). The modest TWFE share increase for Q1 is a denominator artifact of this enrollment decline, not progressive sorting. Balanced-panel estimates restricted to WV colleges with complete data yield null results. West Virginia is uninformative as a test of the sorting mechanism—the institutional context prevents the three-destination mechanism from operating—rather than a contradiction of the Tennessee finding.

8.3 The Decomposition Interpretation

The adoption DiD and the shock-interaction analysis are two empirical manifestations of a single theoretical framework. The adoption DiD captures the *substitution effect* of introducing a merit scholarship: the permanent relative price reduction that generates predictions P1–P7. The shock-interaction analysis captures the *income effect* of economic fluctuations: the cyclical sensitivity of composition to local affordability shocks, which the scholarship may or may not compensate (predictions P8–P11).

The Tennessee finding illustrates both components. HOPE generates a progressive compositional shift on average (substitution effect: $+1.3$ pp for Q_1) but does not stabilize that shift during economic downturns (income effect: zero aggregate buffering, $p = 0.97$). Bruce and Carruthers (2014) document the mechanism: HOPE creates marginal four-year students—low-income families who attend four-year colleges only because the scholarship makes it affordable—who downshift to two-year colleges when family income falls. The scholarship keeps them in the system but does not prevent tier reallocation (Proposition 5).

Florida Bright Futures, by contrast, generates both. The Medallion tier's percentage-based award (75% of tuition) produces progressive sorting and compensates the income effect: the five-for-five insurance gradient shows near-complete neutralization of income shocks across the entire quintile distribution. West Virginia's PROMISE (full tuition coverage) similarly buffers. The distinguishing feature is not threshold height or program generosity in dollars, but whether the

award *scales with costs*—the theoretical distinction formalized in Proposition 4.

This interpretation reconciles an otherwise puzzling pattern. Tennessee and Georgia—the two states with the highest per-student merit-aid expenditures—show the weakest shock buffering. Florida and West Virginia—with lower per-student expenditures—show the strongest. Raw generosity does not determine resilience; award structure does.

8.4 Mobility Implications

Descriptive evidence suggests that the progressive composition shift did not come at the cost of diminished economic returns. Figure 17 shows that upward mobility rates at Tennessee public colleges tracked below the national control average in the pre-HOPE period and converged toward it afterward. Despite substantial access expansion, upward mobility did not decline.[7]

[7] We explored an instrumental variables strategy using a shift-share (Bartik) instrument, but pre-trend tests reveal that the instrument predicts mobility outcomes in the pre-treatment period (joint $F = 20.6$, $p < 0.001$), indicating an exclusion restriction violation. The conservative reading from the descriptive evidence is that colleges whose bottom-quintile representation rose did not experience worse average outcomes.

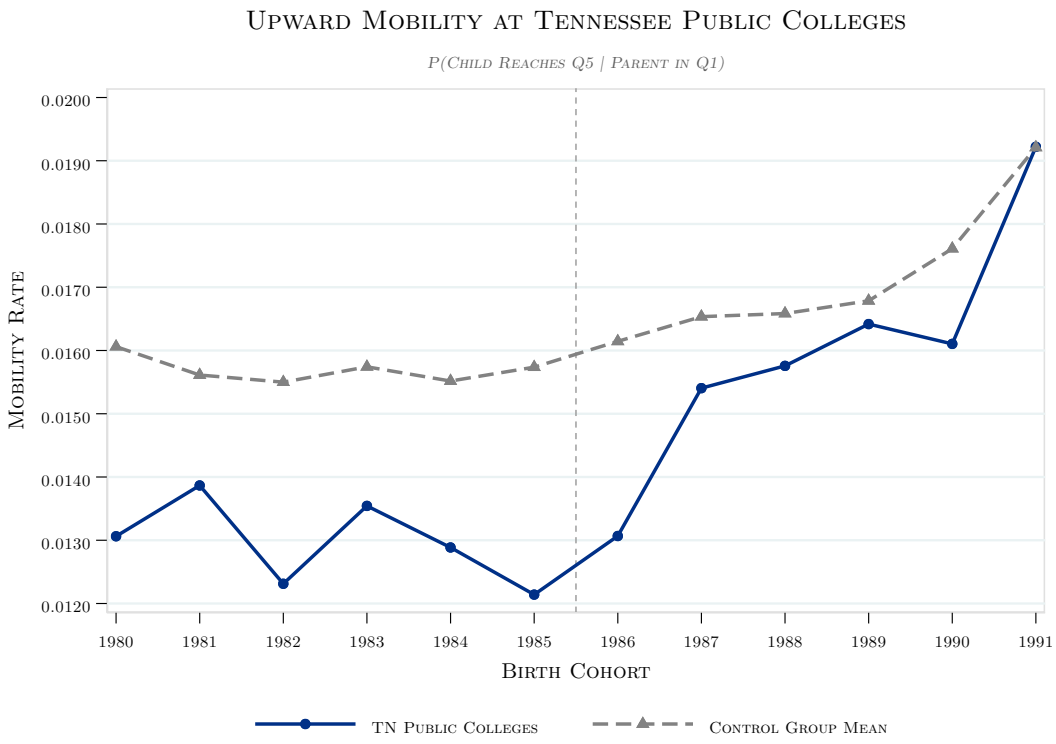

Figure 17: Upward Mobility Rates: Tennessee Public Colleges vs. Control States

Notes: Mobility rate is the probability that a student from the bottom income quintile reaches the top quintile (mr_kq5_pq1). Group 1 = Tennessee public colleges; Group 0 = control-state public colleges. The vertical dashed line marks treatment onset. This is a descriptive comparison, not a causal estimate. Data: Chetty et al. (2020).

8.5 Gender Heterogeneity and the Eligibility Channel

The three-destination model predicts that compositional sorting operates through the price channel: the scholarship S reduces the effective cost of public college for students whose ability exceeds the eligibility threshold τ. We exploit a sharp implication of this mechanism. Tennessee HOPE requires a 3.0 high school GPA. Female students clear this threshold at substantially higher rates than male students—a well-documented consequence of the persistent gender gap in high school academic performance (Goldin et al., 2006).[8] If HOPE's composition effect operates through the eligibility-price channel, the progressive shift should be larger for the gender with a higher eligibility rate.

We test this prediction using gender-specific parental income measures from Chetty et al. (2020).

[8]Nationally, the female high school GPA distribution is shifted rightward by approximately 0.2–0.3 grade points. Among 2004 high school graduates, approximately 40% of female students held GPAs above 3.0, compared to 30% of male students (National Center for Education Statistics, 2023).

We pool Tennessee and Massachusetts as treated states (staggered at $t^* = 1987$ and $t^* = 1988$ respectively) against all remaining states and estimate a stacked interaction model with college-by-gender fixed effects, cohort-by-gender fixed effects, and state-specific linear gender trends.[9]

The female-specific progressive shift in bottom-quintile share is 0.87 pp larger than the male shift (SE = 0.10 pp, $p < 0.001$). To interpret: the overall HOPE effect on Q_1 share is approximately +1.3 pp. The gender decomposition reveals that this average masks substantial heterogeneity—the female-specific effect is roughly +1.7 pp while the male-specific effect is roughly +0.8 pp. Women benefit twice as much as men, consistent with their higher eligibility rate (40% vs. 30% clearing the 3.0 GPA threshold). The gender gap grows over time: +0.75 pp in the first three post-treatment cohorts ($p < 0.001$) and +0.98 pp in later cohorts ($p = 0.002$).

The gender differential concentrates in the bottom-quintile share rather than the mean parental income rank. The par_rank gender gap is −0.16 pp ($p = 0.130$), statistically indistinguishable from zero. This pattern is consistent with the eligibility channel operating primarily at the access margin. Low-income women newly eligible for HOPE face the sharpest change in their effective choice set: without HOPE, many would not attend a four-year public institution; with HOPE, the price drops substantially. Higher-income students attend college regardless of HOPE, so the gender-eligibility mechanism has little traction in the middle and upper quintiles. The resulting composition shift is concentrated in the tail of the income distribution—visible in share measures but diluted in the mean.

This test has a key identification advantage over the main DiD specification. Male and female students attend the same colleges in the same birth cohorts. Any college-level confound—changes in local economic conditions, institutional quality, or admissions policy—affects both genders equally and is absorbed by the college-by-gender fixed effects. Any cohort-level confound—federal financial aid changes, national labor market shifts—is similarly absorbed. For a secular trend to produce the observed gender gap, it would need to differentially shift the income composition of female versus male students at Tennessee public colleges relative to control states, in a pattern that begins precisely at the HOPE treatment date. No plausible alternative satisfies this requirement. Figure 18 displays the gender-specific event studies. Both series track near zero in the pre-period and diverge

[9]State-specific linear gender trends absorb pre-existing differential drift in male versus female composition across states, which is present in the raw data and produces misleading pre-trend violations without detrending. After detrending, the pre-treatment gender interaction is statistically indistinguishable from zero ($p = 0.913$).

after treatment onset, with the female series rising more steeply.

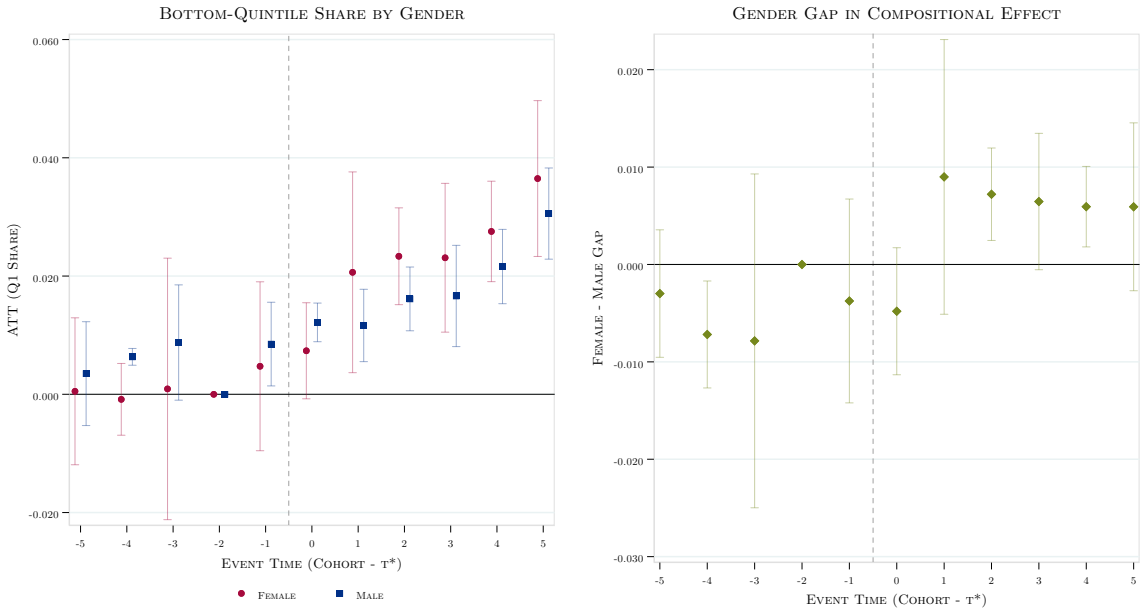

Figure 18: Gender Heterogeneity in HOPE's Compositional Effect

Notes: Callaway–Sant'Anna doubly-robust event-study estimates for bottom-quintile share (Panel A) and mean parental income rank (Panel B), estimated separately for male and female students. The vertical dashed line marks treatment onset ($t^* = 1986$). Reference period $k = -1$. Standard errors clustered at the state level. Gender-specific outcomes from Chetty et al. (2020), Table 5.

The gender channel also connects to the tier reallocation mechanism documented in Section 6.7. The shock-interaction analysis finds that the four-year anti-buffering effect is substantially larger for female students ($\hat{\beta}_2 = -0.128$, $p = 0.005$) than for male students (-0.054, $p = 0.151$). This is the dynamic counterpart of the static gender result: women are more likely to qualify for HOPE, more likely to be the marginal four-year students the scholarship creates, and therefore more sensitive to income shocks at the four-year/two-year margin. The scholarship enables their upward tier substitution in good times and exposes them to downward reallocation in bad times—precisely the income-effect channel formalized in Proposition 5.

The gender result provides our most direct test of the theoretical mechanism. The three-destination model's price channel predicts a specific pattern—larger progressive shifts for the gender with higher eligibility rates—and the data confirm it with high statistical precision. This evidence complements the cross-state comparisons in the following subsections, which test the model's pre-

dictions along the program-design dimension rather than the eligibility dimension.

8.6 Massachusetts Adams Scholarship

The Adams Scholarship (2004) provides tuition waivers at Massachusetts public colleges for students scoring in the top 25% of their district on the MCAS exam (Cohodes & Goodman, 2014). Like Tennessee's HOPE, the subsidy is concentrated at public institutions—private colleges receive no benefit—creating a public-private price differential. The eligibility threshold (top 25% of district) is moderate, qualifying a broad share of the income distribution.

The Callaway–Sant'Anna ATT for bottom-quintile share at Massachusetts public colleges is +1.38 percentage points ($p < 0.001$), remarkably close to Tennessee's +1.27 pp. Mean parental income rank declines by 1.08 percentage points ($p < 0.001$). The Massachusetts result provides independent validation of the Tennessee finding: a second state with a moderate eligibility threshold and a public-sector subsidy concentration produces a progressive compositional shift of similar magnitude.

8.7 South Dakota Opportunity Scholarship

South Dakota's Opportunity Scholarship requires a minimum ACT score of 24 (approximately the 75th percentile nationally) and provides a comparatively small award ($1,000–$5,000 per year). The high threshold and small subsidy satisfy the model's conditions for regressive sorting: only upper-income students are likely to qualify, and the price differential is too small to induce meaningful sector switching.

The Callaway–Sant'Anna ATT confirms this prediction: bottom-quintile share at South Dakota public colleges *declines* by 1.47 percentage points ($p < 0.001$), and mean parental income rank *increases* by 3.25 percentage points ($p < 0.001$). The effect is strongly regressive—the opposite of Tennessee and Massachusetts. The quintile decomposition is also inverted: Q1 through Q3 lose share, Q4 and Q5 gain share.

8.8 Four-State Comparison

Figure 19 displays event-study estimates for bottom-quintile share across all four states. The visual contrast is striking: Tennessee and Massachusetts trend upward after treatment (progressive),

South Dakota trends sharply downward (regressive), and West Virginia shows no clear pattern (null, confounded by enrollment contraction).

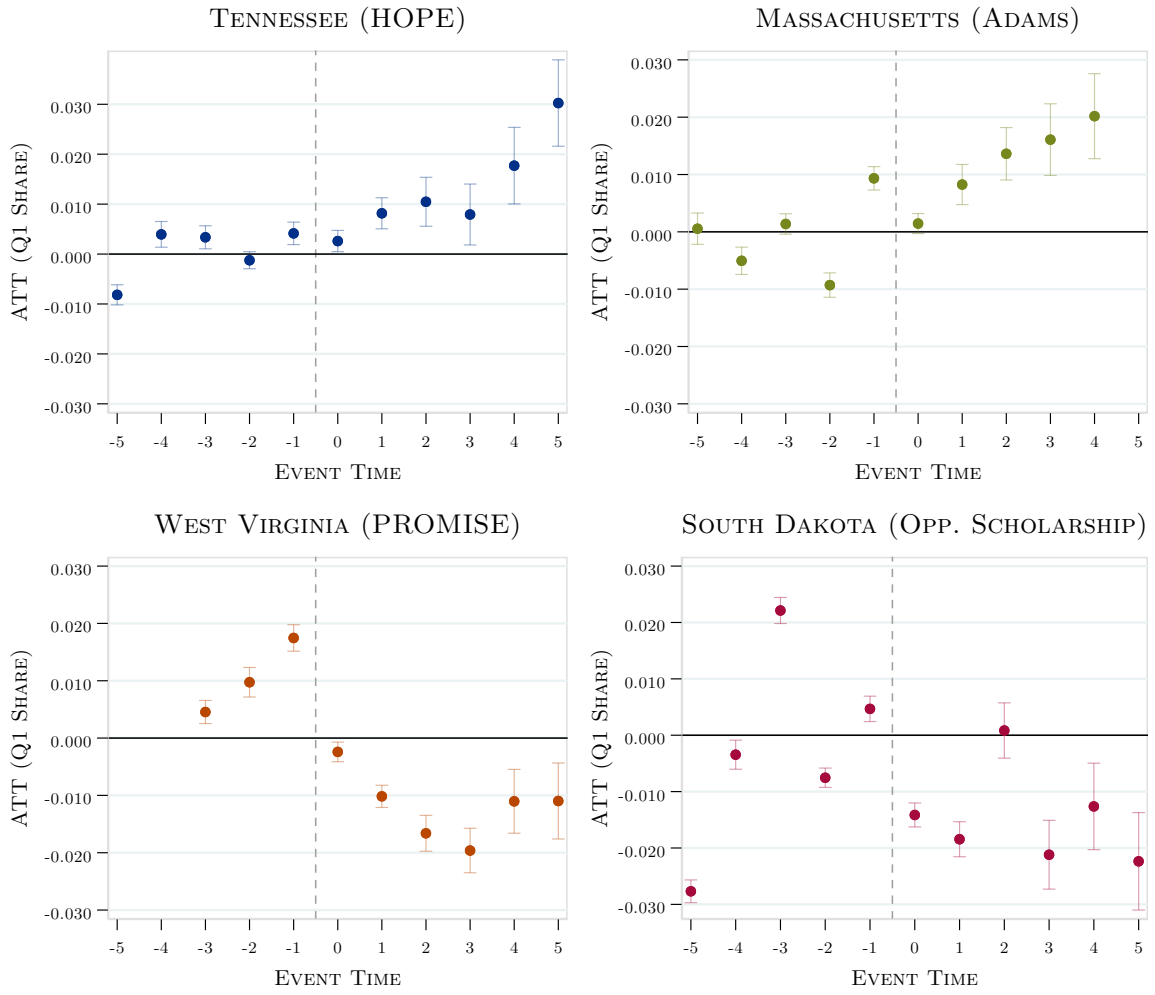

Figure 19: Cross-State Evidence: Merit Aid and Institutional Composition

Notes: Callaway–Sant'Anna event-study estimates for bottom-quintile share at public colleges. Tennessee and Massachusetts show progressive compositional shifts; South Dakota shows a regressive shift; West Virginia shows no significant pattern. Vertical bars denote 95% confidence intervals. Standard errors clustered at the state level. Data: Chetty et al. (2020).

Table 6 maps program design parameters to predicted and observed sorting direction. The three-destination model correctly predicts the direction of compositional change in all four cases where program parameters and institutional context are accounted for.

Table 6: Program Design, Compositional Sorting, and Economic Resilience

State	Program	Threshold	Award Type	Adoption DiD	Shock Buffer
Tennessee	HOPE	3.0 GPA	Fixed $	Progressive	None
Massachusetts	Adams	Top 25%	Tuition waiver	Progressive	Moderate[a]
Florida	Bright Futures	3.0 GPA[b]	% of tuition	— [c]	Strong
West Virginia	PROMISE	3.0 GPA	Full tuition	Null[†]	Strong
Georgia	HOPE	3.0 GPA	Reduced	Regressive	None
South Dakota	Opp. Schol.	ACT 24+	Fixed $	Regressive	None

Notes: "Adoption DiD" reports the direction of the bottom-quintile share effect from the program adoption difference-in-differences. "Shock Buffer" reports whether the program buffers composition during local income downturns (Section 6.6). [a]Massachusetts shows significant buffering on par_rank ($p = 0.008$) but not Q_1 ($p = 0.14$). [b]Medallion Scholars tier (operative for broad coverage). [c]Florida's program predates the Chetty panel; no adoption DiD is estimable. [†]West Virginia's null adoption result reflects enrollment contraction. Its shock-buffer result (+0.096, $p = 0.024$) is the strongest among all programs.

8.9 Generalizability and Policy Implications

The cross-state evidence extends beyond a single treated state. Massachusetts independently produces a progressive shift of similar magnitude under similar program parameters, while South Dakota's regressive shift confirms the model's prediction under high thresholds. The shock-insurance analysis adds Florida and West Virginia as new testable cases—states whose programs predate the Chetty panel and were previously untestable through adoption DiD.

The three-destination model, extended with the Slutsky decomposition, organizes all cases through a common set of design parameters: eligibility threshold (determines *who* is eligible), public-private subsidy differential (determines *where* students sort), Pell interaction (determines *whether* low-income students benefit), and award structure (determines *whether* progressive sorting persists during economic fluctuations). No single parameter is sufficient; the combination determines both the direction and durability of compositional change.

For policymakers designing merit-aid programs to promote low-income access, the findings suggest four principles. First, the eligibility threshold must be low enough that substantial numbers of low-income students qualify (moderate GPA, OR rather than AND conditions). Second, the sub-

sidy must be concentrated at public institutions to create a meaningful price differential. Third, the program should not offset existing need-based aid (no Pell displacement). Fourth—and this is the new contribution of the shock-insurance analysis—the award should scale with tuition costs (percentage-based or full-tuition coverage) rather than provide a fixed dollar amount. Fixed-dollar awards generate progressive sorting but leave it vulnerable to economic downturns; tuition-scaling awards generate progressive sorting *and* sustain it.

The WV non-replication highlights an additional scope condition: program design is necessary but not sufficient. Enrollment contraction at WV public colleges overwhelmed compositional dynamics in the adoption DiD, though WV PROMISE shows the strongest shock-buffering effect among all programs. A stable or growing public system with absorptive capacity is required for the sorting mechanism to operate.

9 Conclusion

We document that following Tennessee's HOPE Scholarship, public colleges experienced a progressive shift in socioeconomic composition. The share of bottom-quintile students increased by 1.3 percentage points, mean parental income rank declined by 0.8–1.5 percentage points (depending on estimator), and the effect is monotonically ordered across all five income quintiles. Lower-income students gained representation while higher-income students lost it.

This compositional change reflects sorting across sectors rather than enrollment expansion. Total enrollment at Tennessee public colleges is unchanged post-HOPE. The level decomposition confirms that the number of bottom-quintile students increased while the number of top-quintile students decreased. Top-quintile students who exit the public sector do not appear at Tennessee private colleges, consistent with out-of-state enrollment as the relevant margin.

We decompose this compositional effect into a permanent substitution effect—the relative price reduction that generates progressive sorting—and a cyclical income effect—the sensitivity of that sorting to local economic conditions. Merging BEA county income data with the Chetty panel, we show that the income effect is compensated by tuition-scaling programs (Florida Bright Futures, West Virginia PROMISE) but not by fixed-dollar programs (Tennessee HOPE, Georgia HOPE). The shock-insurance analysis produces a second five-for-five monotonic quintile gradient, estimated

on entirely different variation, consistent with the adoption-DiD gradient. Award structure—not threshold height or raw generosity—determines whether progressive sorting is sustained during economic downturns.

The cross-state evidence supports the model's comparative statics. Massachusetts' Adams Scholarship independently produces a progressive shift of similar magnitude. South Dakota's Opportunity Scholarship produces a strongly regressive shift—as the model predicts. Florida and West Virginia, whose programs predate the Chetty panel and are not testable through adoption DiD, are testable through the shock-insurance design and show the strongest buffering effects. The three-destination model, extended with the Slutsky decomposition, organizes all cases through a common set of design parameters.

Four limitations merit emphasis. First, the analysis relies on ecological inference from college-level aggregates. Second, with 26 state-level clusters, inference requires supplementation with permutation tests. Third, while the preferred specification improves Rambachan–Roth sensitivity substantially (Δ_{RM} robust at $\bar{M} \leq 1$, Δ_{SD} robust at $M \leq 0.005$), the smoothness restriction remains the binding constraint at moderate bounds. The shock-insurance results provide independent corroboration through different identifying variation. Fourth, the shock-interaction analysis shows that most individual-state estimates are statistically insignificant (only Florida and West Virginia are significant on Q_1); the power limitation reflects the 12-cohort panel and within-state shock variation.

For policy, the findings suggest four principles for merit-aid program design. First, set the eligibility threshold low enough that a substantial share of lower-income students qualifies. Second, concentrate the subsidy at public institutions to create a meaningful sector price differential. Third, do not offset existing need-based aid. Fourth, structure the award as a percentage of tuition rather than a fixed dollar amount, so that progressive sorting is sustained when family budgets tighten. The relevant counterfactual is not whether merit aid is more progressive than need-based aid—it almost certainly is not—but whether merit aid *at the margin* improves the composition and resilience of the institutions that serve the most students.

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A Figures

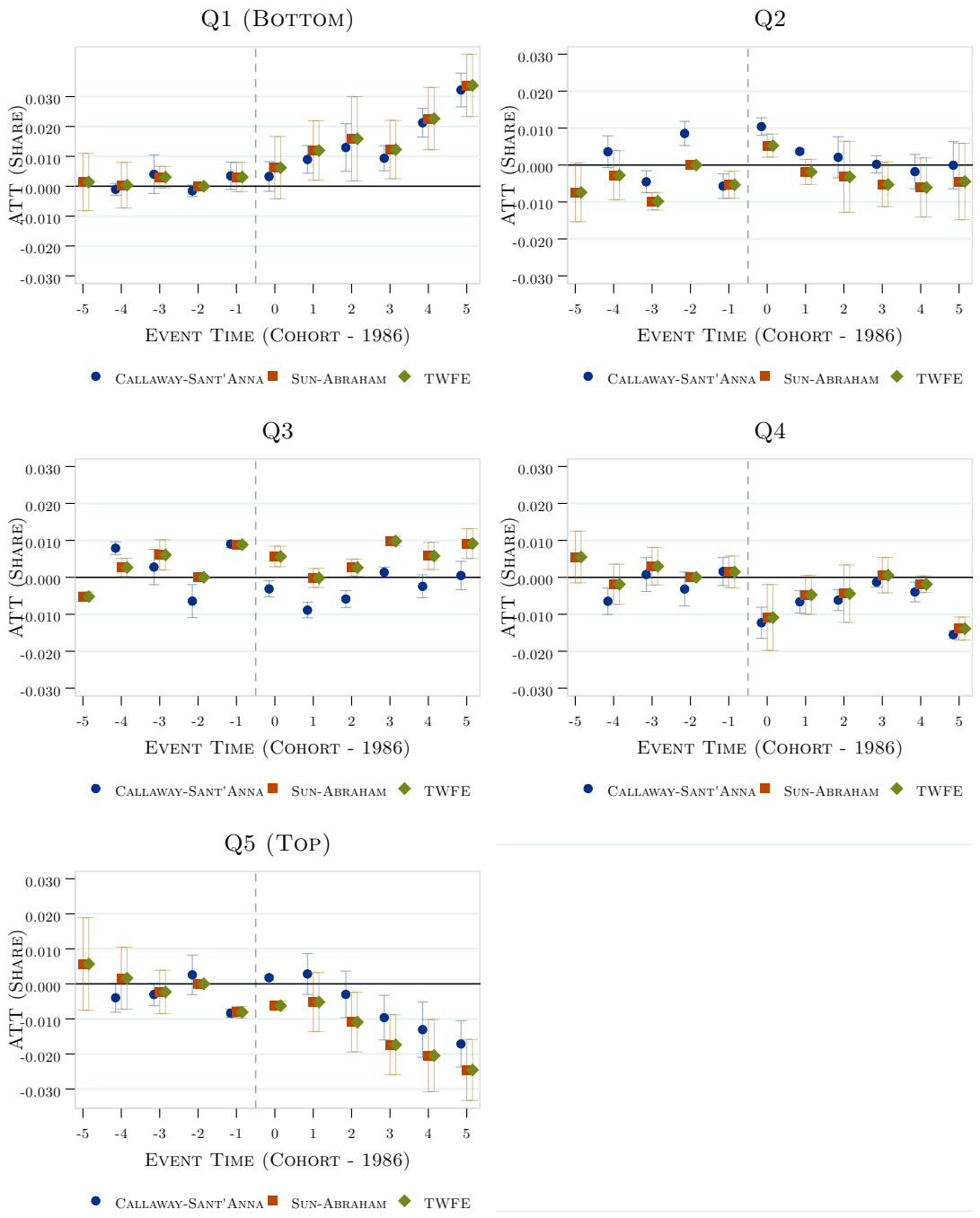

Figure A1: Event-Study Estimates by Income Quintile

Notes: Event-study estimates from three heterogeneity-robust estimators (Callaway and Sant'Anna (2021), Sun and Abraham (2021), and TWFE), estimated separately for each income quintile share. Bottom quintiles (Q1–Q2) trend upward post-treatment (gaining share); top quintiles (Q4–Q5) trend downward (losing share); Q3 is near zero. The monotonic gradient is consistent across all three estimators. Bordering-state control group (AL, NC, VA). $k = -2$ omitted (1984 cohort, last untreated). 1980 birth cohort dropped. Standard errors clustered by state. Data: Chetty et al. (2020).

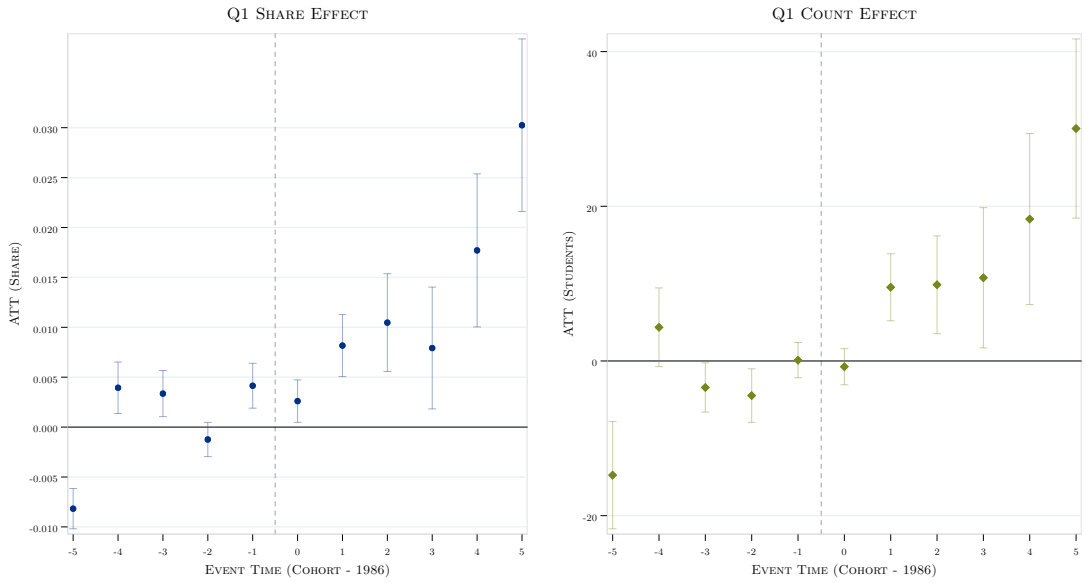

Figure A2: Share Effects vs. Count Effects: Bottom-Quintile Students

Notes: Comparison of share-based (left) and count-based (right) event-study estimates for bottom-quintile students. The share increase is accompanied by a count increase, confirming that the composition shift reflects genuine sorting rather than a denominator artifact. Count outcomes are inherently noisier than share outcomes because they inherit enrollment variation across colleges (coefficient of variation ≈ 1.0). The wider confidence intervals on count estimates reflect this additional noise, not weaker identification. Callaway and Sant'Anna (2021) estimator. Data: Chetty et al. (2020).

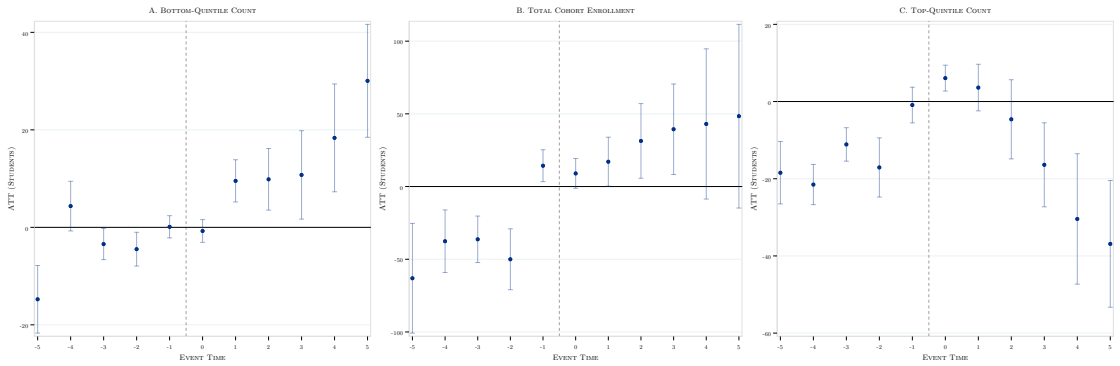

Figure A3: Level Decomposition: Bottom-Quintile Count, Top-Quintile Count, and Total Enrollment

Notes: Callaway and Sant'Anna (2021) event-study estimates for count outcomes (quintile share × cohort enrollment). The critical finding is that total enrollment (Panel B) is unchanged post-treatment, ruling out the denominator-artifact explanation for share-based results. Q1 count increases and Q5 count decreases, consistent with real compositional sorting. Count outcomes exhibit wider confidence intervals than share outcomes because they compound share-level variation with enrollment variation (enrollment coefficient of variation ≈ 1.0 across colleges). The pre-treatment noise in count event studies reflects this inherent volatility rather than a failure of parallel trends for the share-based specifications. Data: Chetty et al. (2020).

Figure A4: Spillover Test: Tennessee Private Colleges

Notes: Callaway and Sant'Anna (2021) event-study estimates comparing Tennessee private colleges (treated) to private colleges in control states. No significant composition change is detected, indicating that top-quintile students who exit Tennessee public colleges do not re-sort to in-state private institutions. $N = 5{,}336$. Data: Chetty et al. (2020).

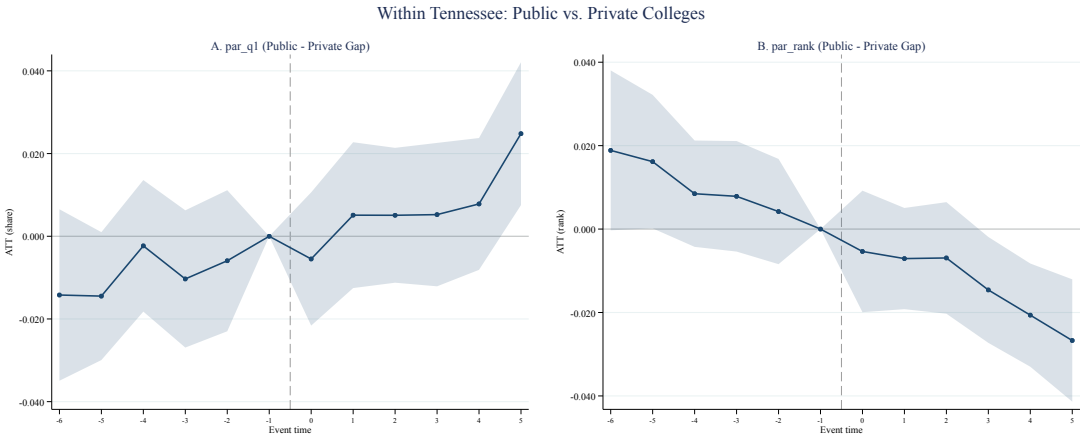

Figure A5: Within-Tennessee Comparison: Public vs. Private Colleges

Notes: Event-study estimates restricting the sample to Tennessee institutions only. Public colleges (treated) are compared to Tennessee private colleges (control). The composition shift is concentrated in the public sector. College and cohort fixed effects included. $N = 571$. Data: Chetty et al. (2020).

B Tables

Table A1: Distribution of HOPE Scholarship Recipients by Sector, Fall 2004–2008

	2004	2005	2006	2007	2008
Panel A: Number of Recipients					
TBR Universities	7,454	7,054	7,637	7,956	8,121
TBR Community Colleges	4,506	3,909	4,720	4,996	4,970
UT Campuses	5,380	5,310	5,576	5,988	6,327
Private Colleges	3,109	3,387	3,492	3,885	3,902
Total	**20,449**	**19,660**	**21,425**	**22,825**	**23,320**
Panel B: Share of Recipients (%)					
TBR Universities	36	36	36	35	35
TBR Community Colleges	22	20	22	22	21
UT Campuses	26	27	26	26	27
Private Colleges	15	17	16	17	17

Notes: First-time freshmen receiving Tennessee HOPE Scholarship awards. TBR = Tennessee Board of Regents; UT = University of Tennessee system. Panel B shares may not sum to 100 due to rounding. Public institutions (TBR + UT) account for 83–85% of recipients in all years. Source: Tennessee Student Assistance Corporation (Tennessee Higher Education Commission, 2025).

Table A2: Difference-in-Differences: Pre- and Post-Treatment Means

	Treated (TN Public)	**Control** (Non-Merit States)
Panel A: Pre-Treatment Means		
Bottom-quintile share	0.138	0.120
Panel B: Post-Treatment Means		
Bottom-quintile share	0.159	0.129
Panel C: DiD Estimate		
Treated × Post	0.0148^{***}	
	(0.0016)	
College FE	✓	
Cohort FE	✓	
Observations	23,395	
R^2	0.948	
Clusters	27	

Notes: Standard errors clustered at the state level in parentheses. Treated = Tennessee public colleges. Control = public colleges in 26 states without broad-based merit programs (Sjoquist & Winters, 2015). Post = birth cohorts 1985 and later. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A3: Callaway–Sant'Anna Aggregate ATT Estimates

	(1) Bottom-Quintile Share	(2) Parental Income Rank
ATT	0.0126^{***}	-0.0081^{***}
	(0.0023)	(0.0018)
95% CI	[0.0080, 0.0171]	[−0.0117, −0.0046]

Notes: Aggregate average treatment effect on the treated from Callaway and Sant'Anna (2021). Preferred specification: $t^* = 1987$, 1980 cohort dropped, BEA county income control. The comparison group consists of never-treated public colleges in 26 non-merit-program states. Analytical standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A4: Static TWFE Difference-in-Differences Estimates

	(1) Bottom-Quintile Share	(2) Parental Income Rank
TN Public × Post	0.0154^{***}	-0.0185^{***}
	(0.0030)	(0.0025)
Constant	0.138^{***}	0.540^{***}
	(0.0001)	(0.0001)
Observations	6,289	6,289
Adjusted R^2	0.935	0.974
Clusters	27	27

Notes: Two-way fixed effects estimates with college and cohort fixed effects. Standard errors clustered at the state level in parentheses. Treated = Tennessee public colleges. Post = birth cohort $\geq$ 1986. The sample includes public colleges in Tennessee and 26 non-merit-program states. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A5: Quintile Decomposition: TWFE Static DiD by Income Quintile

	(1)	(2)	(3)	(4)	(5)
	Q1 (Bottom)	Q2	Q3	Q4	Q5 (Top)
TN Public × Post	0.0142***	0.0071***	0.0032*	−0.0091***	−0.0154***
	(0.0029)	(0.0023)	(0.0017)	(0.0018)	(0.0019)
Observations	5,630	5,630	5,630	5,630	5,630

Notes: Each column reports the TWFE static difference-in-differences estimate from a separate regression in which the dependent variable is the share of students from the indicated income quintile. Preferred specification: $t^* = 1987$, 1980 cohort dropped, BEA county income control. College and cohort fixed effects included. Standard errors clustered at the state level in parentheses. Quintile shares sum to one by construction. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A6: Level Decomposition: Share and Count Outcomes

	(1)	(2)	(3)	(4)	(5)
	Q1 Share	Q1 Count	Total Count	Q5 Share	Q5 Count
TN Public × Post	0.0154***	11.23**	−2.11	−0.0179***	−34.38***
	(0.0030)	(4.256)	(25.42)	(0.0025)	(8.635)
Constant	0.138***	167.9***	1,586***	0.200***	449.7***
	(0.0001)	(0.104)	(0.622)	(0.0001)	(0.211)
Observations	6,289	6,289	6,289	6,289	6,289
Adjusted R^2	0.935	0.948	0.985	0.978	0.989
Clusters	27	27	27	27	27

Notes: Two-way fixed effects estimates with college and cohort fixed effects. Standard errors clustered at the state level in parentheses. Count outcomes are constructed as quintile share × cohort size. The Q1 share increase (column 1) is accompanied by a Q1 count increase (column 2), while total enrollment is unchanged (column 3), confirming that share changes reflect compositional sorting rather than a denominator artifact. Post = birth cohort ≥ 1986. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A7: Heterogeneity by Institution Tier: Two-Year vs. Four-Year Colleges

	Two-Year			Four-Year		
Outcome	Coef.	SE	N	Coef.	SE	N
Bottom-quintile share	0.0159^{***}	(0.0034)	3,632	0.0084^{***}	(0.0022)	1,985
Parental income rank	-0.0124^{***}	(0.0025)	3,632	-0.0180^{***}	(0.0017)	1,985
Q1 count	3.11	(7.89)	3,632	22.19^{***}	(6.50)	1,985
Q5 count	-16.80^{**}	(6.84)	3,632	-36.04^{**}	(17.33)	1,985
Total count	-48.80	(45.63)	3,632	146.07^{***}	(39.23)	1,985

Notes: TWFE static difference-in-differences estimates estimated separately for two-year (community colleges) and four-year institutions. Preferred specification: $t^* = 1987$, 1980 cohort dropped, BEA county income control. College and cohort fixed effects included. Standard errors clustered at the state level in parentheses. Two-year colleges show displacement (composition shifts, enrollment unchanged); four-year colleges show expansion (enrollment grows, composition shifts). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A8: Spillover Test: Tennessee Private Colleges

Outcome	Coef.	SE	p-value	N
Bottom-quintile share	0.0001	(0.0011)	0.914	5,010
Parental income rank	-0.0043^{**}	(0.0018)	0.023	5,010

Notes: TWFE static DiD comparing Tennessee private colleges to private colleges in non-merit-program control states. Preferred specification: $t^* = 1987$, 1980 cohort dropped. The null effect on bottom-quintile share indicates that top-quintile students exiting Tennessee public colleges do not re-sort to in-state private institutions. Standard errors clustered at state level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A9: Within-Tennessee Comparison: Public vs. Private Colleges

Outcome	Coef.	SE	p-value	N
Bottom-quintile share	0.0156^{***}	(0.0037)	< 0.001	537
Parental income rank	-0.0193^{***}	(0.0033)	< 0.001	537

Notes: TWFE static difference-in-differences restricted to Tennessee institutions only. Treated = Tennessee public colleges; control = Tennessee private colleges. Preferred specification: $t^* = 1987$, 1980 cohort dropped. College and cohort fixed effects included. Robust standard errors in parentheses. The small sample reflects the within-state restriction ($N = 537$). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A10: Robustness Summary Across Estimators, Timing, Control Groups, and Sensitivity

	Bottom-Quintile Share		Parental Income Rank	
Specification	Coef.	SE	Coef.	SE
Panel A: Estimator (preferred spec: $t^* = 1987$, drop 1980, +BEA)				
C&S	0.0126^{***}	(0.0023)	-0.0081^{***}	(0.0018)
TWFE static	0.0142^{***}	(0.0040)	-0.0148^{***}	(0.0033)
Panel B: Treatment Timing (TWFE, national, drop 1980, +BEA)				
$t^* = 1986$	0.0135^{***}	(0.0041)	-0.0159^{***}	(0.0035)
$t^* = 1987$ (preferred)	0.0142^{***}	(0.0040)	-0.0148^{***}	(0.0033)
Panel C: Control Group (TWFE, $t^* = 1987$, drop 1980, +BEA)				
National (preferred)	0.0142^{***}	(0.0040)	-0.0148^{***}	(0.0033)
Bordering (AL, VA, NC)	0.0124^{***}	(0.0023)	-0.0104^{***}	(0.0027)
Panel D: BEA Controls (TWFE, $t^* = 1987$, national, drop 1980)				
Without BEA	0.0158^{***}	(0.0043)	-0.0184^{***}	(0.0038)
With BEA level (preferred)	0.0142^{***}	(0.0040)	-0.0148^{***}	(0.0033)
With BEA level + change	0.0132^{***}	(0.0041)	-0.0155^{***}	(0.0035)
Panel E: Rambachan–Roth Sensitivity (preferred spec)				
Original CI	[0.007, 0.013]		[−0.009, −0.003]	
Δ_{RM}: robust at $\bar{M} \leq$	1.0		0.5	
Δ_{SD}: robust at $M \leq$	0.005		0.005	

Notes: Panels A–D report static difference-in-differences estimates under alternative specifications. All specifications include college and cohort fixed effects. Standard errors clustered at the state level in parentheses. The preferred specification uses the national control group, drops the 1980 birth cohort, sets $t^* = 1987$, and controls for BEA county log per capita income at decision year (cohort +17). Panel E summarizes Rambachan and Roth (2023) sensitivity results for Q_1 under the preferred specification: the original CI excludes zero; Δ_{RM} is robust at $\bar{M} \leq 1.0$ (post-treatment violations as large as the maximum pre-treatment violation); Δ_{SD} is robust at $M \leq 0.005$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

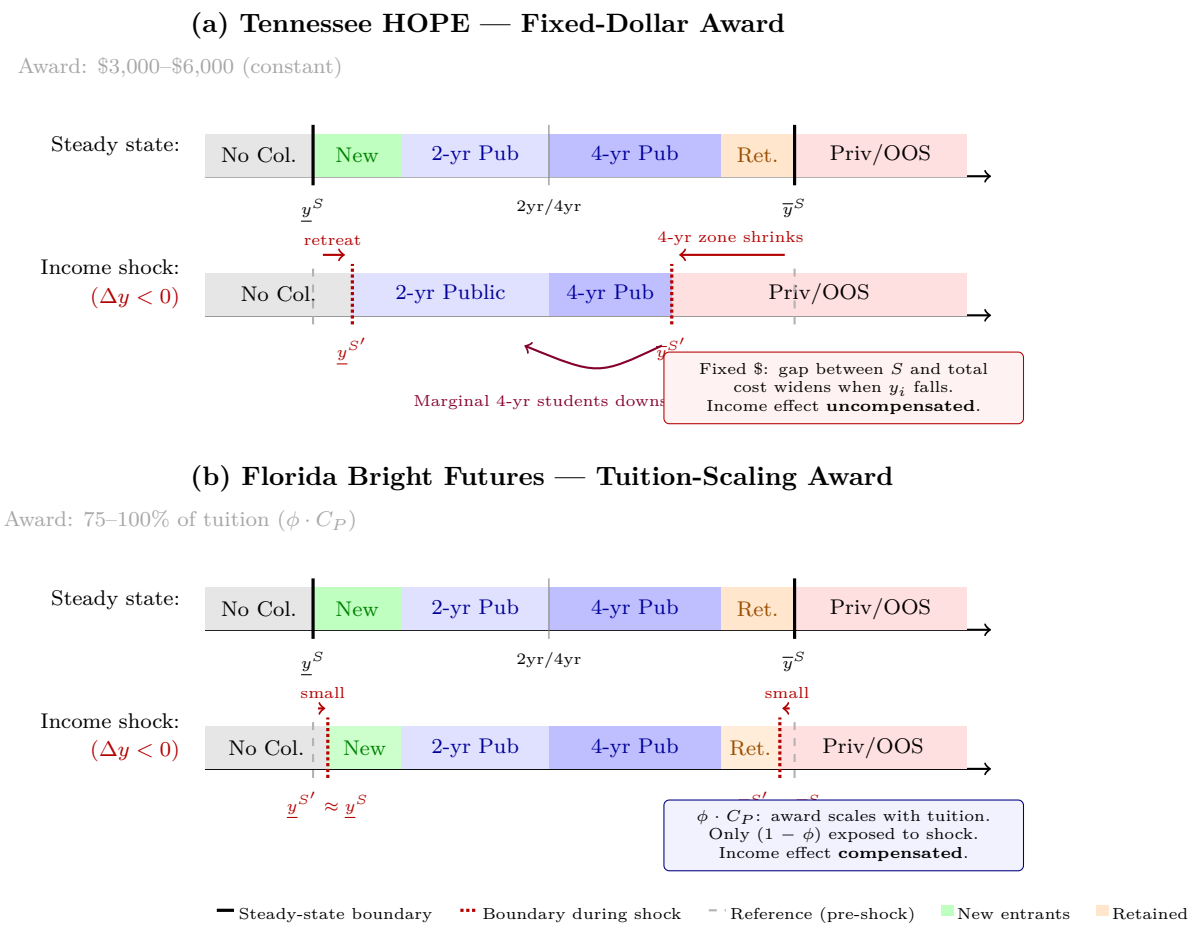

Figure 2: Slutsky Decomposition: How Award Structure Determines Resilience to Income Shocks

Notes: Each panel shows the steady-state sorting (top bar) and the response to a negative income shock (bottom bar). Panel (a): Tennessee HOPE pays a fixed dollar amount ($3,000–$6,000). When income falls, the gap between the award and total cost of attendance widens. Both boundaries retreat substantially: some new entrants at the access margin drop out, and marginal four-year students (created by HOPE) downshift to two-year colleges. The income effect is uncompensated. Panel (b): Florida Bright Futures covers a fixed percentage of tuition (75–100%). When income falls, the award scales with tuition costs, so the gap between scholarship and cost barely changes. Boundaries shift minimally. The income effect is compensated. Both programs produce progressive sorting in steady state (the substitution effect); only the percentage-based program sustains it during downturns.

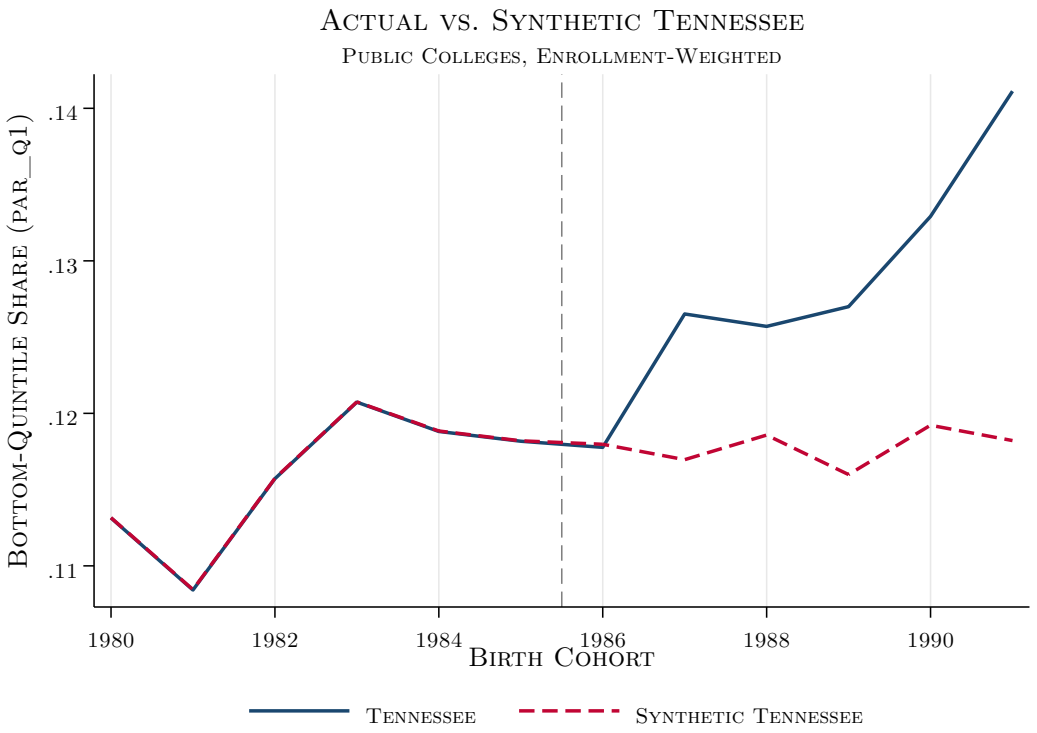

Figure 3: Actual vs. Synthetic Tennessee: Bottom-Quintile Share at Public Colleges

Notes: Synthetic Tennessee is constructed from a weighted combination of 22 non-merit states using the method of Abadie et al. (2010), matching on pre-treatment bottom-quintile share (1980–1984). The vertical dashed line indicates the first HOPE-eligible birth cohort. The two series track closely in the pre-treatment period, then diverge after HOPE implementation. Data: Chetty et al. (2020).

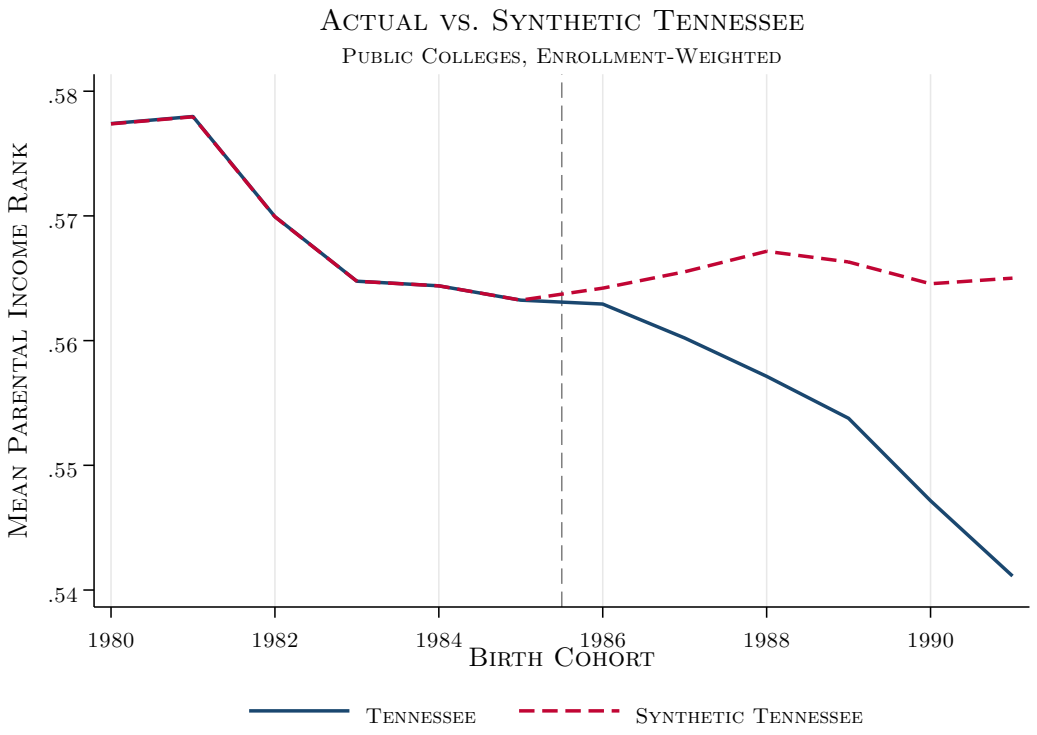

Figure 4: Actual vs. Synthetic Tennessee: Mean Parental Income Rank at Public Colleges

Notes: Synthetic Tennessee constructed as in Figure 3, matching on pre-treatment mean parental income rank (1980–1984). Tennessee's actual parental income rank diverges downward after HOPE implementation, consistent with the progressive compositional shift documented in the bottom-quintile share results. Data: Chetty et al. (2020).

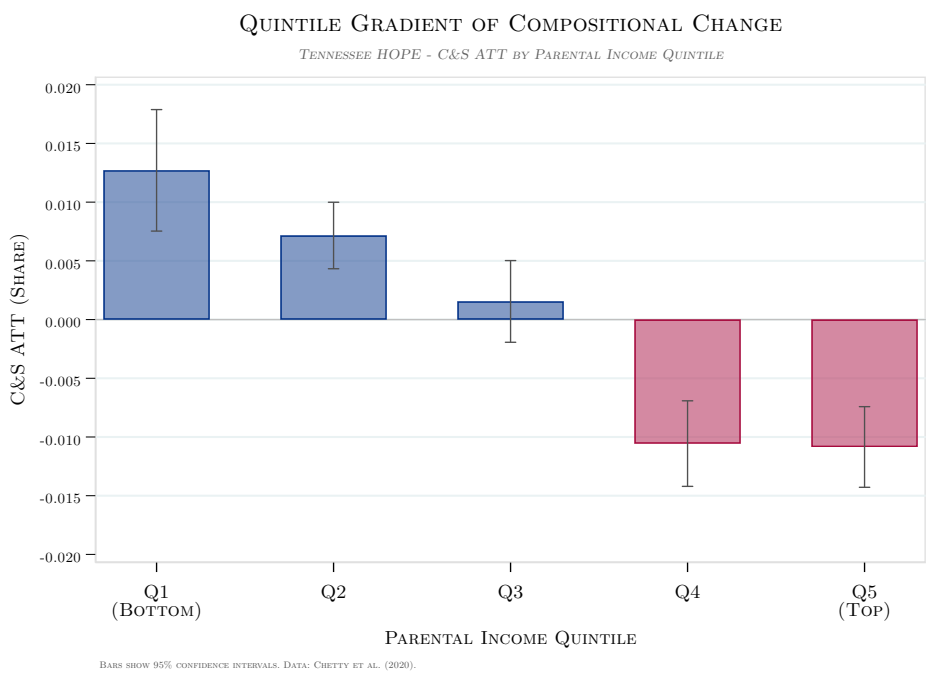

Figure 5: Quintile Gradient of Compositional Change Following Tennessee HOPE

Notes: Callaway and Sant'Anna (2021) ATT estimates for each parental income quintile share. The effect is monotonically ordered: bottom-quintile students gain the most representation and top-quintile students lose the most. Significance is strongest at the tails (Q1, Q5) where both the theoretical effect and cross-college variance are largest. Bars denote 95% confidence intervals. Data: Chetty et al. (2020).

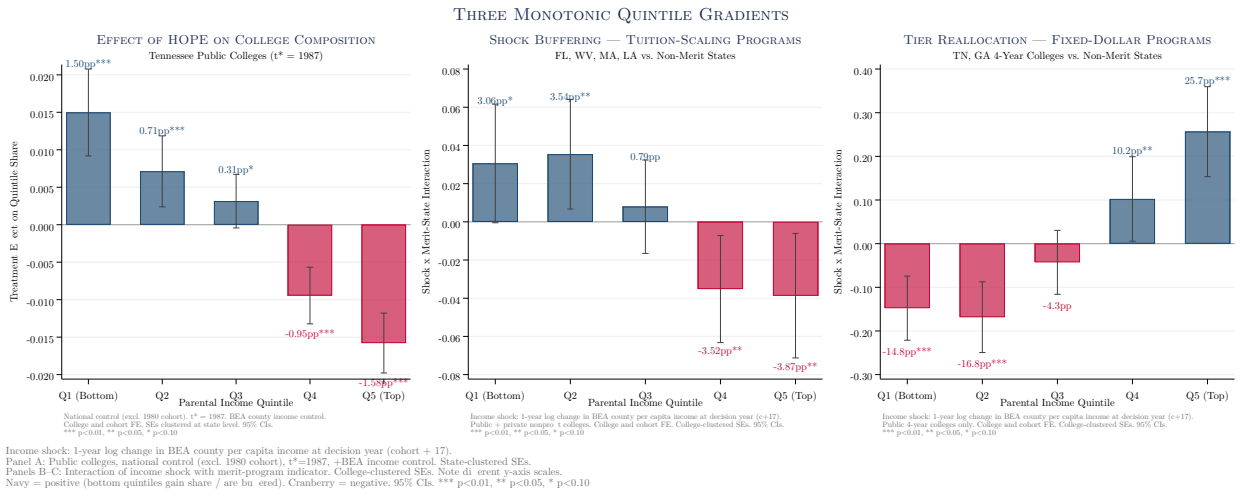

Figure 8: Three Monotonic Quintile Gradients

Notes: Three independent quintile gradients. Panel A: adoption DiD (preferred specification). Panel B: shock-insurance interaction (tuition-scaling programs × county income shock). Panel C: shock-substitution interaction (fixed-dollar programs × county income shock at four-year public colleges). All three exhibit monotonic ordering across all five quintiles. Panels B and C use BEA county per capita personal income change at decision year (cohort + 17). 95% CIs shown. See text for specifications.

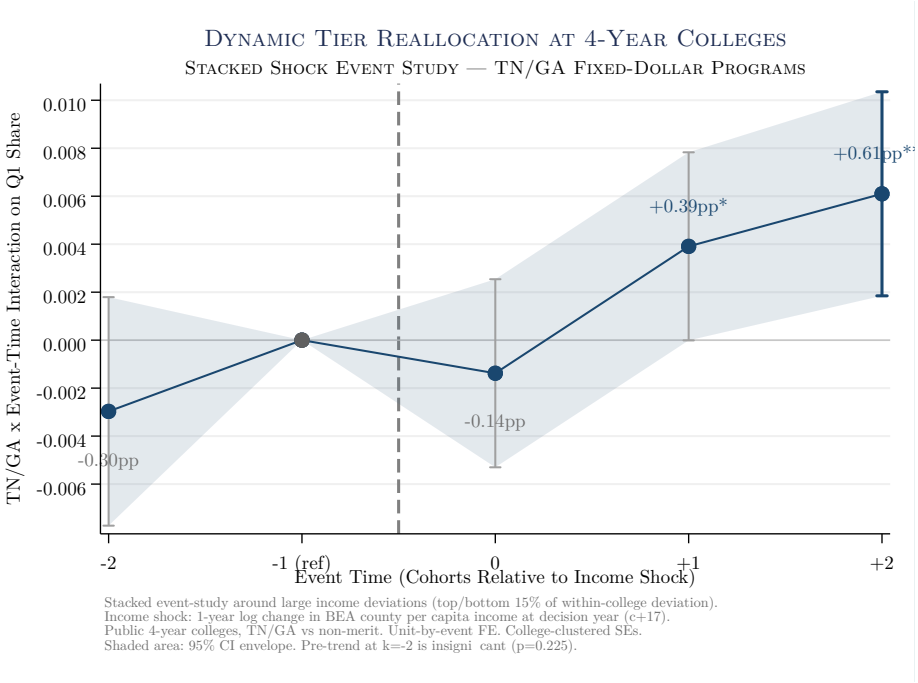

Figure 9: Dynamic Tier Reallocation at Four-Year Colleges

Notes: Stacked shock-event-study around large income deviations (top/bottom 15% of within-college deviation from mean). Shows the TN/GA × event-time interaction coefficient on Q_1 share at four-year public colleges. The pre-trend at $k = -2$ is insignificant ($p = 0.225$). Composition displaces on impact then rebounds by $k = +2$ (+0.61 pp, $p = 0.005$). Unit-by-event FE, college-clustered SEs. Shaded area: 95% CI envelope.

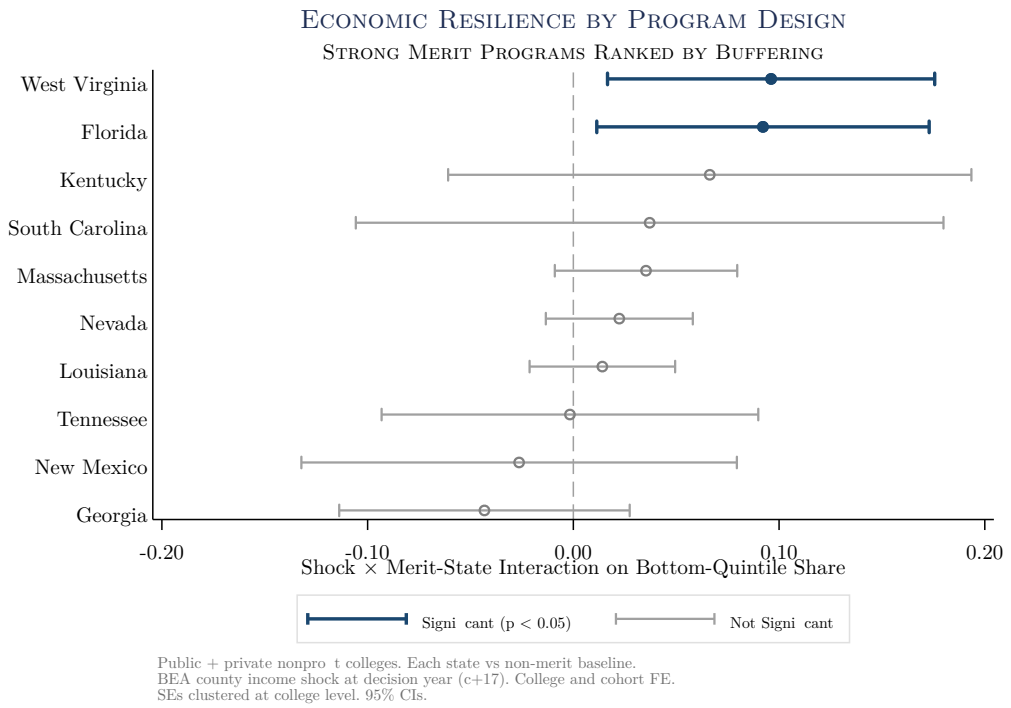

Figure 10: Economic Resilience by Program Design

Notes: Shock × merit-state interaction coefficient on bottom-quintile share for each strong merit program (Sjoquist & Winters, 2015, plus Massachusetts). Each state estimated separately against non-merit baseline. Public and private nonprofit colleges. BEA county income shock at decision year (cohort + 17). College and cohort FE, college-clustered SEs. 95% CIs.